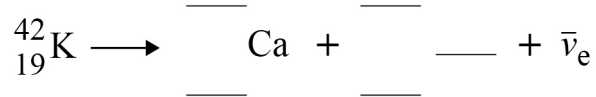


|   |   |
|---|---|
| 0 | 1 |
|---|---|

Complete the equation for the radioactive decay of  ${}^{42}_{19}\text{K}$ .

**[2 marks]**

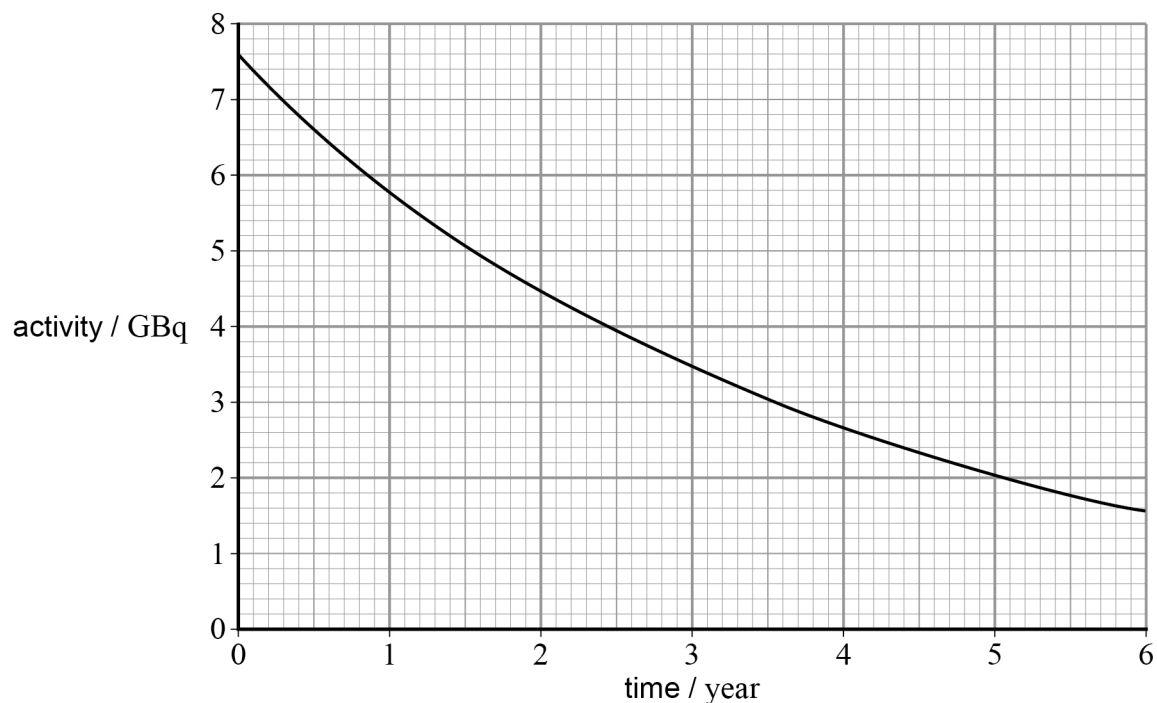


|       |
|-------|
| <hr/> |
| 2     |

07

Promethium-147 ( $^{147}_{61}\text{Pm}$ ) is a radioactive nuclide that emits  $\beta^-$  particles.  
**Figure 6** shows the variation with time of the activity of a  $^{147}_{61}\text{Pm}$  source.

**Figure 6**



07

1

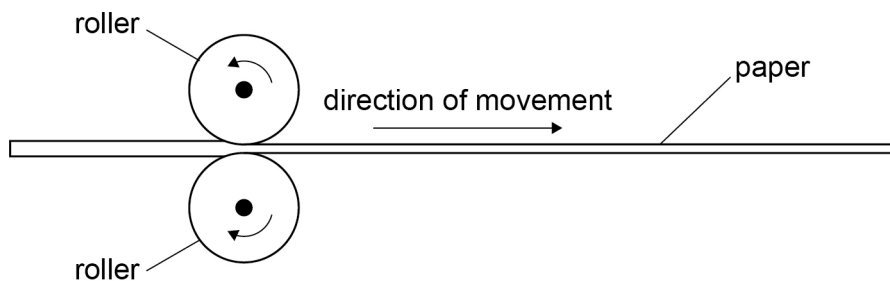
Determine, using **Figure 6**, an accurate value for the half-life of  $^{147}_{61}\text{Pm}$ .

**[3 marks]**

half-life = \_\_\_\_\_ year



### Figure 7



Describe how  $^{147}_{61}\text{Pm}$  is used in this process and the radioactive properties that make  $^{147}_{61}\text{Pm}$  suitable for this purpose.

**[4 marks]**

[illegible]



1 0

Time-lapse photography can be used to show the changing positions of an object at regular time intervals. The different positions are displayed in a single photograph.

A ball is dropped from rest and its motion is analysed using time-lapse photography to determine a value for the acceleration due to gravity,  $g$ .

**Figure 11** shows a full-scale representation of a time-lapse photograph of the ball as it falls. The position of the ball is shown from time  $t = 0$  and then at 40 ms intervals.

1 0 . 1

Show, on **Figure 11**, the distance you would measure to give an accurate value for  $g$ . Label this distance  $s$ .

[1 mark]

1 0 . 2

Justify your answer to question 10.1

[1 mark]

---

---

---

---

1 0 . 3

Measure distance  $s$  using a ruler with a mm scale.

[1 mark]

$s =$  \_\_\_\_\_



**Figure 11**

- $t = 0$
- $t = 40 \text{ ms}$
- $t = 80 \text{ ms}$
- $t = 120 \text{ ms}$
- $t = 160 \text{ ms}$
- $t = 200 \text{ ms}$

**Question 10 continues on the next page**



10.4

Determine, using  $s = \frac{1}{2} g t^2$  and your value for  $s$ , a value for  $g$ .

[1 mark]

$g =$  \_\_\_\_\_  $\text{m s}^{-2}$

10.5

Estimate the percentage uncertainty in your measurement of  $s$ .

[1 mark]

percentage uncertainty in  $s =$  \_\_\_\_\_ %

10.6

The percentage uncertainty in  $t$  is 0.8%

Use this and your values from question 10.4 and question 10.5 to calculate the absolute uncertainty in your value for  $g$ .

[3 marks]

absolute uncertainty in  $g = \pm$  \_\_\_\_\_  $\text{m s}^{-2}$

Turn to page 20 for the next question



**Turn over for the next question**

*Do not write  
outside the  
box*

**DO NOT WRITE ON THIS PAGE  
ANSWER IN THE SPACES PROVIDED**

**Turn over ►**



1 1 . 1 Define impulse.

[1 mark]

---



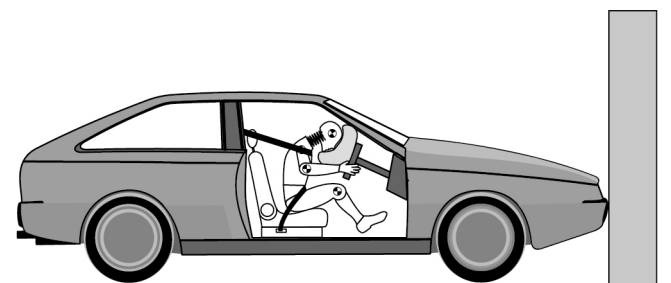
---



---

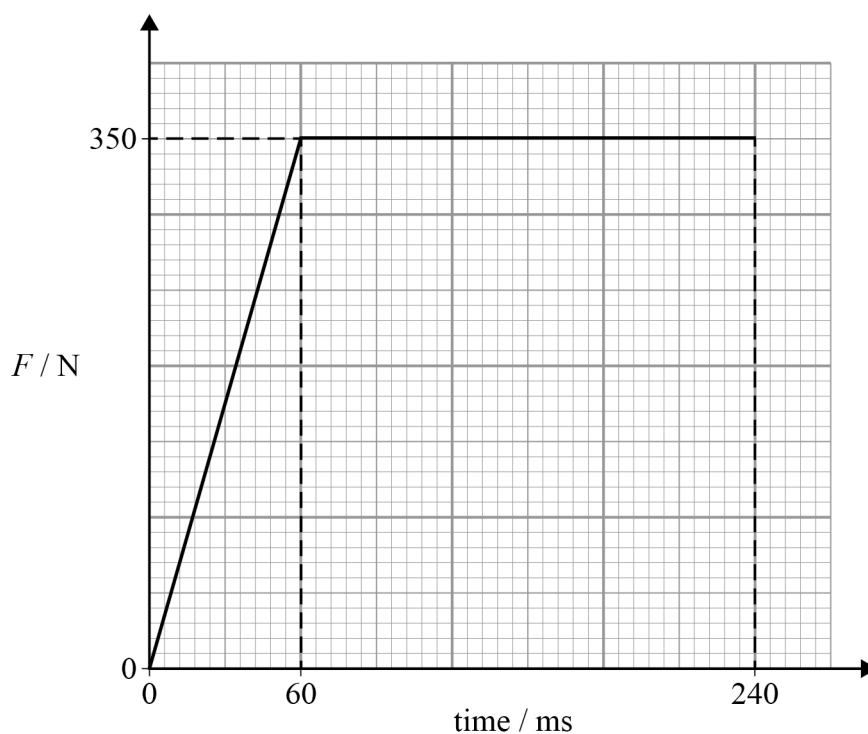
1 1 . 2 A vehicle-safety engineer investigates airbags. A car has a dummy in the driver's seat and crashes into a wall so that the airbag inflates as shown in **Figure 12**. The force  $F$  experienced by the dummy's head is monitored.

**Figure 12**



**Figure 13** is a graph showing the variation of  $F$  with time from when the head first touches the airbag until it stops moving.

**Figure 13**



Show that the impulse experienced by the head is approximately  $75 \text{ N s}$

[3 marks]

1 1 . 3 The dummy's head has a mass of  $4.8 \text{ kg}$

Calculate the initial velocity of the head.

[2 marks]

velocity = \_\_\_\_\_  $\text{m s}^{-1}$

1 1 . 4 In other safety tests car **A**, without a crumple zone, hits the wall at  $12 \text{ m s}^{-1}$  and rebounds at  $8 \text{ m s}^{-1}$   
Car **B** has a crumple zone. It also hits the wall at  $12 \text{ m s}^{-1}$  but does not rebound.

State **two** reasons why a dummy in car **A** would be likely to be more damaged than a dummy in car **B**.

[2 marks]

---

---

---

---

---

---

---



**Section C**

Each of Questions **12** to **25** is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional sheets for this working.

**1 2** Which list contains symbols for SI fundamental base units only?

[1 mark]

**A** A, kg, m, N, s

☐

**B** A, K, m, mol, s

☐

**C** C, kg, m, mol, rad

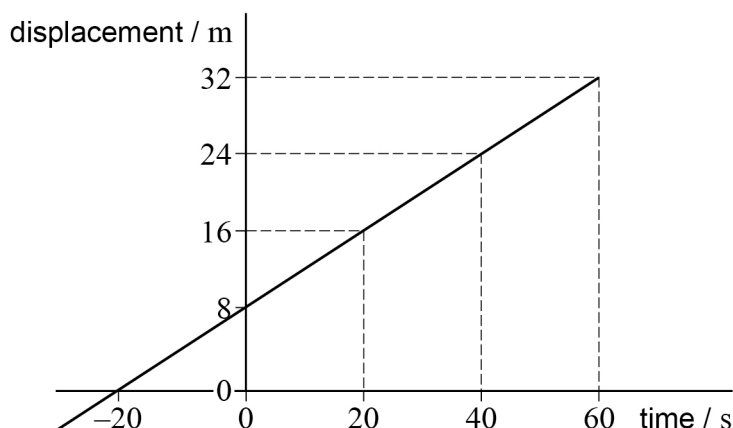
☐

**D** J, K, kg, m, s

☐

**1 3**

The graph shows the variation with time of the displacement of a moving vehicle from a fixed point.



What are the velocity and displacement of the moving vehicle 20 s after it passes the fixed point?

**[1 mark]**

|          | Velocity / $\text{m s}^{-1}$ | Displacement / m |                       |
|----------|------------------------------|------------------|-----------------------|
| <b>A</b> | 0.53                         | 8                | <input type="radio"/> |
| <b>B</b> | 0.40                         | 8                | <input type="radio"/> |
| <b>C</b> | 0.53                         | 16               | <input type="radio"/> |
| <b>D</b> | 0.40                         | 16               | <input type="radio"/> |

**1 4**

Which list contains the names of vector quantities only?

**[1 mark]**

**A** acceleration, force, momentum, power, weight

☐

**B** acceleration, displacement, power, velocity, weight

☐

**C** displacement, force, momentum, velocity, weight

☐

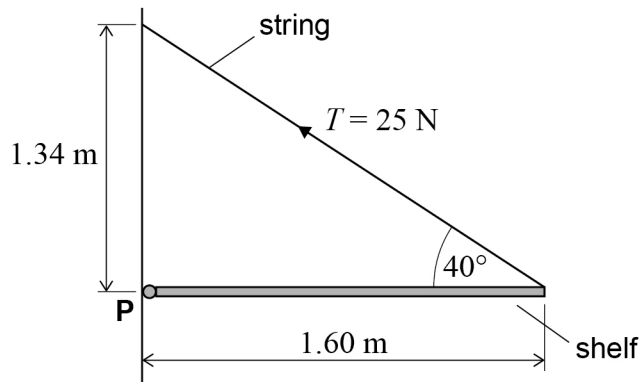
**D** displacement, length, momentum, weight, work done

☐

**1 5**

The diagram shows a shelf, hinged at **P** and supported by a string. The tension  $T$  in the string is 25 N

Do not write  
outside the  
box



What is the moment of  $T$  about **P**?

[1 mark]

**A**  $1.34 \times 25 \tan 40$

☐

**B**  $1.60 \times 25 \cos 40$

☐

**C**  $1.34 \times 25 \sin 40$

☐

**D**  $1.60 \times 25 \sin 40$

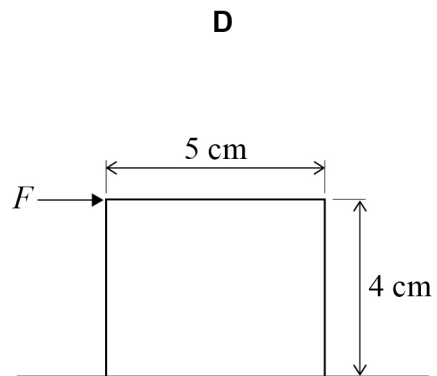
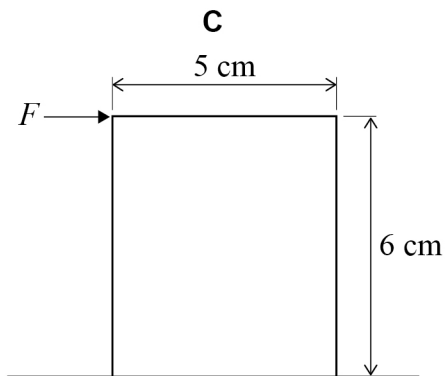
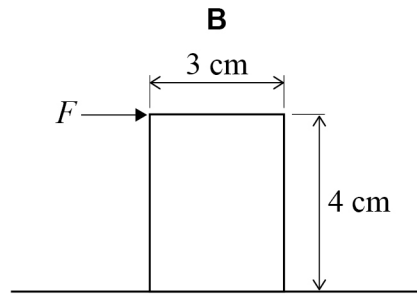
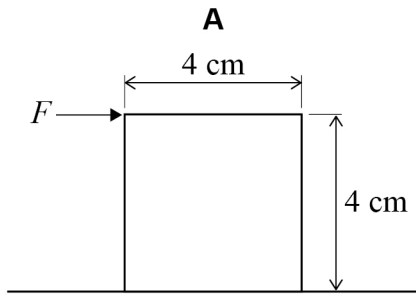
☐

1 6

Do not write  
outside the  
box

The solid objects below all have the same weight. Each object experiences a horizontal force on its top edge. None of the objects slide along the surface.

Which object requires the smallest force to cause it to move?



[1 mark]

- A** ☐
- B** ☐
- C** ☐
- D** ☐

Turn over for the next question

Turn over ►



- 1 7** A moving train accelerates uniformly at  $0.70 \text{ m s}^{-2}$  up to a speed of  $55 \text{ m s}^{-1}$ .  
The train travels 1875 m during this acceleration.

What are the average speed of the train and the time taken for the acceleration?

[1 mark]

|          | Average speed of the train / $\text{m s}^{-1}$ | Time taken / s |                       |
|----------|------------------------------------------------|----------------|-----------------------|
| <b>A</b> | 37.5                                           | 50             | <input type="radio"/> |
| <b>B</b> | 37.5                                           | 107            | <input type="radio"/> |
| <b>C</b> | 48.2                                           | 19             | <input type="radio"/> |
| <b>D</b> | 48.2                                           | 138            | <input type="radio"/> |

- 1 8** Two spheres of equal mass, **P** and **Q**, are dropped and fall until reaching their terminal speeds.  
**P** has a larger diameter than **Q**.

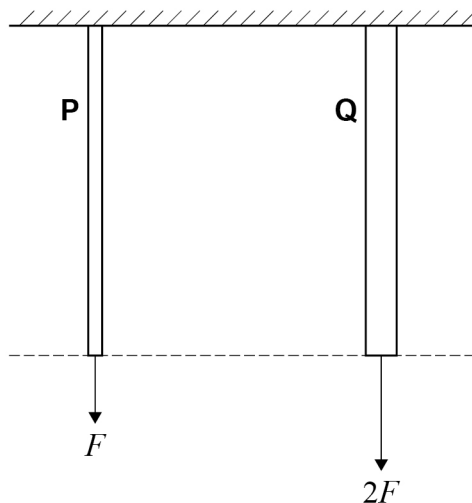
Which line describes the magnitudes of **Q**'s initial acceleration and terminal speed compared with those for **P**?

[1 mark]

|          | Initial acceleration of Q | Terminal speed of Q       |                       |
|----------|---------------------------|---------------------------|-----------------------|
| <b>A</b> | less than for <b>P</b>    | greater than for <b>P</b> | <input type="radio"/> |
| <b>B</b> | less than for <b>P</b>    | same as for <b>P</b>      | <input type="radio"/> |
| <b>C</b> | same as for <b>P</b>      | same as for <b>P</b>      | <input type="radio"/> |
| <b>D</b> | same as for <b>P</b>      | greater than for <b>P</b> | <input type="radio"/> |



Two wires, **P** and **Q**, have the same original length. They are stretched so that they have the same extensions.



The tensile force in **P** is  $F$  and the tensile force in **Q** is  $2F$ .

**P** has a diameter  $d$  and **Q** has a diameter  $2d$ .

**P** is made from a material of Young modulus  $E$ .

What is the Young modulus of the material from which **Q** is made?

[1 mark]

**A**  $\frac{E}{4}$

☐

**B**  $\frac{E}{2}$

☐

**C**  $E$

☐

**D**  $2E$

☐

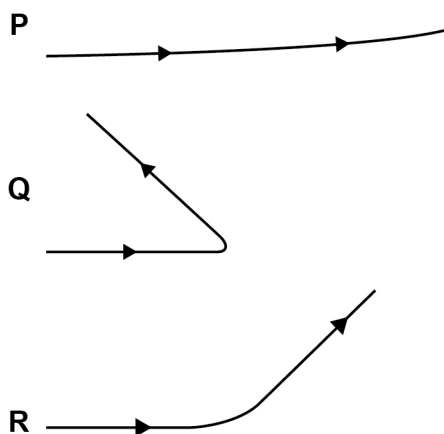
Turn over for the next question

Turn over ►



2 0

**P**, **Q** and **R** are the paths of alpha particles in Rutherford's alpha-scattering experiment. All of the alpha particles have the same initial energy and the paths are drawn to the same scale.



Which row identifies the order of the paths starting with the smallest distance of closest approach and ending with the greatest distance of closest approach to the target nucleus?

[1 mark]

**A** P, Q, R

☐

**B** Q, R, P

☐

**C** Q, P, R

☐

**D** R, P, Q

☐

2 1

What is the specific charge of an alpha particle relative to the specific charge of a proton?

[1 mark]

**A**  $\frac{1}{4}$

☐

**B**  $\frac{1}{2}$

☐

**C** 2

☐

**D** 4

☐


**2 2**

$^{207}\text{Pb}$  can be formed from a series of decays beginning with  $^{211}\text{Pb}$ . Only alpha particles and beta particles are emitted in the series.

How many alpha and beta particles are emitted in the series?

**[1 mark]**

**A** 1 alpha particle and 1 beta particle

☐

**B** 1 alpha particle and 2 beta particles

☐

**C** 2 alpha particles and 1 beta particle

☐

**D** 2 alpha particles and 2 beta particles

☐**2 3**

A detector is placed 50 cm from a gamma source that has a half-life of 40 s  
The corrected count rate is found to be  $1400 \text{ count s}^{-1}$

What will be the corrected count rate 4.0 minutes later when the detector is 2.0 m from the gamma source?

**[1 mark]**

**A**  $1.37 \text{ count s}^{-1}$

☐

**B**  $2.73 \text{ count s}^{-1}$

☐

**C**  $5.47 \text{ count s}^{-1}$

☐

**D**  $10.9 \text{ count s}^{-1}$

☐**2 4**

What are the products of the radioactive decay of a free neutron?

**[1 mark]**

**A** an antineutron and two gamma rays

☐

**B** a proton, an antiproton and an antineutrino

☐

**C** a proton, an electron and an antineutrino

☐

**D** a proton, a positron and a neutrino

☐

**2****5**

Which statement about hazards from radioactive sources is **not** correct?

**[1 mark]**

**A** Exposure times should be minimised by keeping sources in lead boxes when not being used.

☐

**B** A person cannot be contaminated from a sealed source of beta radiation if they are more than 1 m from the source.

☐

**C** Radioactive sources are kept in lead boxes because alpha, beta and gamma radiation cannot penetrate lead.

☐

**D** Radioactive sources should always be handled with tongs to prevent direct contamination of the skin.

☐

*Do not write  
outside the  
box*

0 5

Fluorine-18 ( $^{18}_9\text{F}$ ) is a  $\beta^+$  emitter used in medicine. Its half-life is 110 minutes.

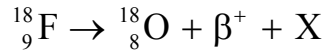
Fluorine-18 decays to oxygen-18 ( $^{18}_8\text{O}$ ) which is stable.

A fluorine-18 compound is injected into a patient and gamma radiation arising from the decay is detected outside the body.

A sample of pure  $^{18}_9\text{F}$  is prepared and then stored for 220 minutes.

0 5 . 1

The decay equation for  $^{18}_9\text{F}$  is



Identify X.

[2 marks]

---

0 5 . 2

Calculate the ratio  $\frac{\text{number of } ^{18}_9\text{F} \text{ nuclei in the sample}}{\text{number of } ^{18}_8\text{O} \text{ nuclei in the sample}}$  after 220 minutes.

[2 marks]

ratio = \_\_\_\_\_

0 5 . 3

Explain why the  $^{18}_9\text{F}$  sample must be used soon after it is produced.

[2 marks]

---

---

---

---



**Turn over for the next question**

*Do not write  
outside the  
box*

**DO NOT WRITE ON THIS PAGE  
ANSWER IN THE SPACES PROVIDED**

**Turn over ►**

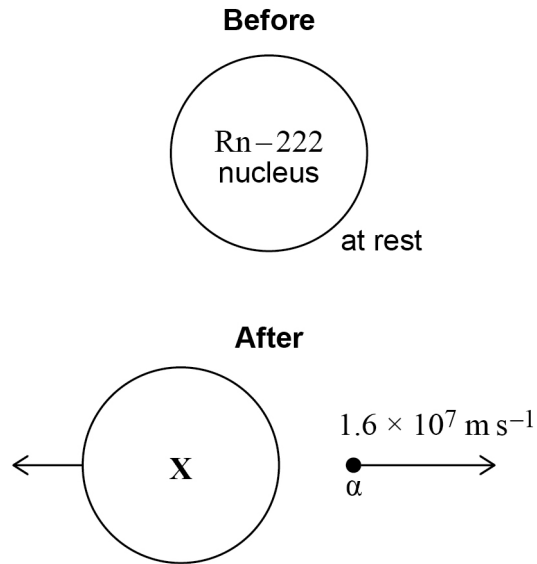


|             |             |
|-------------|-------------|
| Don't write | Don't write |
|-------------|-------------|

A stationary radon-222 ( $\text{Rn-222}$ ) nucleus decays and emits an  $\alpha$  particle of energy 5.5 MeV.

This causes the daughter nucleus **X** to move in the opposite direction as shown in **Figure 3**.

**Figure 3**



The  $\alpha$  particle has mass  $6.6 \times 10^{-27} \text{ kg}$ .

Show that the initial speed of the  $\alpha$  particle is approximately  $1.6 \times 10^7 \text{ m s}^{-1}$ .

**[3 marks]**



**0 7 . 2** Calculate the recoil speed of X.

mass of X =  $3.6 \times 10^{-25}$  kg

**[2 marks]**

recoil speed = \_\_\_\_\_ m s<sup>-1</sup>

**0 7 . 3** When scientists studied beta decay they observed that

- the beta particle and the daughter nucleus did not travel in opposite directions
- beta particles from the decay of a particular nuclide had a range of energies.

Explain how these observations led to the prediction of a previously undiscovered particle.

**[3 marks]**

---

---

---

---

---

---

---

**Turn over for the next question**



|             |             |
|-------------|-------------|
| Don't write | Don't write |
|-------------|-------------|

|   |   |
|---|---|
| 1 | 1 |
|---|---|

A student measures the count rate from a gamma source to determine its half-life.

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

State and explain **one** precaution the student should take to limit his exposure to ionising radiation.

**[2 marks]**

Precaution \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Explanation \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

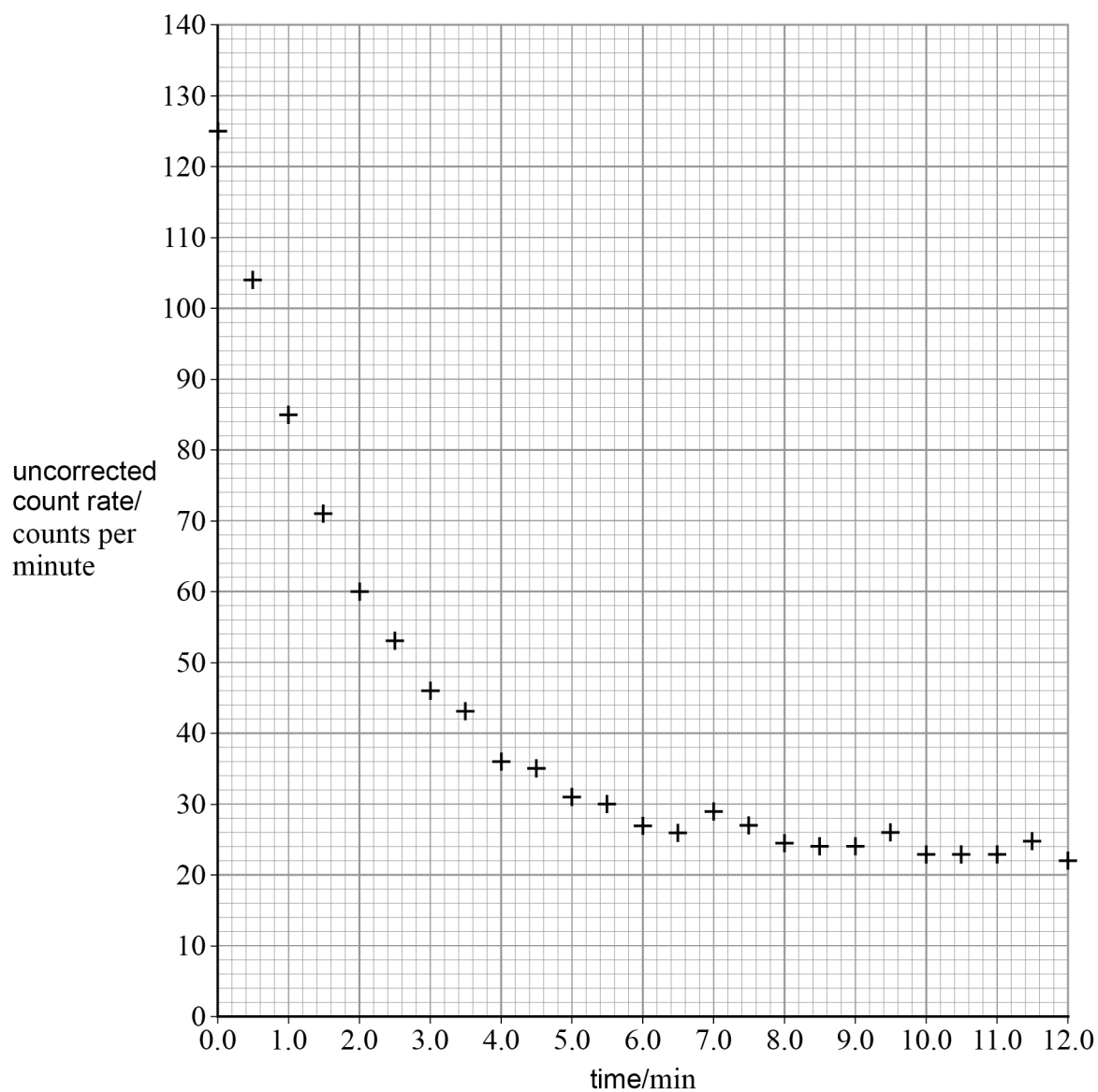
**Question 11 continues on the next page**

**Turn over ►**



1 1 . 2 Figure 8 is a plot of the results.

Figure 8



Estimate the background count rate.

[1 mark]

background count rate = \_\_\_\_\_ counts per minute



**1 1 . 3** Determine the half-life of the source.

**[3 marks]**

half-life = \_\_\_\_\_ min

**1 1 . 4** The student decides to measure the background count rate directly.

Describe how he should do this.

In your answer, explain how he should ensure the accuracy in his measurement.

**[3 marks]**

---

---

---

---

---

---

---

---

---

---

**END OF SECTION B**



## Section C

Each of Questions **12** to **25** is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional sheets for this working.

A Geiger counter is used to measure the corrected count rate  $C$  from a gamma source in a vacuum.

What is the relationship between  $C$  and the distance  $x$  between the Geiger counter and the source?

[1 mark]

**A**  $C$  is directly proportional to  $x^2$

☐

**B**  $C$  is directly proportional to  $x$

☐

**C**  $C$  is inversely proportional to  $x$

☐

**D**  $C$  is inversely proportional to  $x^2$

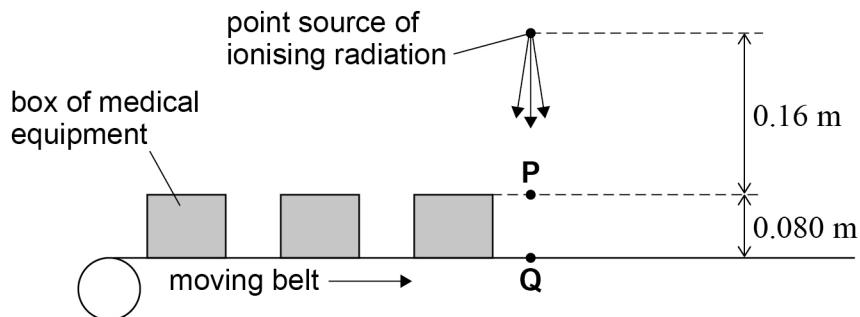
☐

1 2

Medical equipment can be sterilised using ionising radiation. Any bacteria that absorb the ionising radiation will be killed.

Cardboard boxes of equipment to be sterilised are put on a moving belt and passed through a beam of ionising radiation from a point source as shown in **Figure 10**.

**Figure 10**



The boxes are 0.080 m high and the top of each box is 0.16 m below the source.

1 2

1

Suggest **one** advantage of using gamma radiation as the point source in this process.

**[1 mark]**

---



---

1 2

2

High-energy electrons that behave like  $\beta^-$  particles can also be used as the point source in this process.

Suggest **one** advantage of using high-energy electrons in this process.

**[1 mark]**

---



---



1 2 . 3

One manufacturer uses gamma from a cobalt-60 source. **P** and **Q** are points vertically below the point source. For a new source, the gamma radiation intensity at **Q** is  $4.0 \times 10^{-2} \text{ W m}^{-2}$  when there is no box under the source.

Calculate the gamma radiation intensity at **P**.

[2 marks]

gamma radiation intensity = \_\_\_\_\_  $\text{W m}^{-2}$

Question 12 continues on the next page



1 2 . 4

Cobalt-60 has a half-life of 5.3 years. The sterilization process is ineffective if the gamma ray intensity is less than  $8.0 \times 10^{-3} \text{ W m}^{-2}$ .

During which range of times will the intensity of the gamma radiation from the source fall to  $8.0 \times 10^{-3} \text{ W m}^{-2}$ ?

Tick **one** box.

Explain the reason for your answer.

[3 marks]

0–5.3 years

☐

5.3–10.6 years

☐

10.6–15.9 years

☐

15.9–21.2 years

☐

explanation \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

1 2 . 5

A different manufacturer uses high-energy electrons to sterilise the boxes of equipment. When the boxes have passed under the source, they are turned upside down and then passed under the source again.

Suggest why they are turned upside down and passed under the source again.

[1 mark]

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_



## Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional sheets for this working.

The half-life of a radioactive isotope is 2.0 hours.

The count rate measured from a freshly prepared sample of the isotope is 140 counts per minute.

The background count rate is 20 counts per minute.

What will be the measured count rate after 4.0 hours?

[1 mark]

**A** 30 counts per minute

☐

**B** 35 counts per minute

☐

**C** 50 counts per minute

☐

**D** 55 counts per minute

☐

A detector is placed 20 cm from a source of gamma rays.  
The measured count rate is  $C$ .  
The background count rate is  $B$ .

What measured count rate is detected 60 cm from the source?

[1 mark]

**A**  $\frac{C + 8B}{9}$

☐

**B**  $\frac{C + 10B}{9}$

☐

**C**  $\frac{C + 2B}{3}$

☐

**D**  $\frac{C + 4B}{3}$

☐

Turn over for the next question

Turn over ►

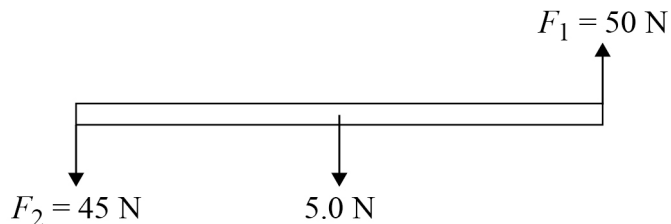


Do not write

**2 6**outside the  
box

The forces  $F_1$  and  $F_2$  are in a vertical plane and are maintained at  $90^\circ$  to the uniform bar shown in the diagram.

The bar has a weight of 5.0 N.



The bar will rotate anticlockwise.

After rotating by less than  $90^\circ$  the bar will also accelerate

**[1 mark]**

**A** downwards and to the left.

☐

**B** downwards and to the right.

☐

**C** upwards and to the left.

☐

**D** upwards and to the right.

☐

[illegible]

[illegible]

[illegible]

[illegible]

|   |   |
|---|---|
| 0 | 1 |
|---|---|

State **two** precautions for the safe use of a gamma source in a school laboratory.

[2 marks]

1 \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

2 \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

|   |
|---|
| 2 |
|---|

|   |   |   |   |
|---|---|---|---|
| 0 | 2 | . | 1 |
|---|---|---|---|

State what is meant by the rest energy of a particle.

[1 mark]

\_\_\_\_\_  
\_\_\_\_\_

|   |   |   |   |
|---|---|---|---|
| 0 | 2 | . | 2 |
|---|---|---|---|

Calculate, in MeV, the rest energy of an electron.

[2 marks]

rest energy = \_\_\_\_\_ MeV

|   |
|---|
| 3 |
|---|



0 3

A tennis ball of mass 58 g has a rough surface.

A cricket ball of mass 160 g has a smooth surface.

Assume that the balls have the same diameter.

Both balls are dropped from the roof of a tall building at the same time and they fall through the air to the ground.

Sketch, on the axes below, graphs to show the variations with time  $t$  of the speed of each ball.

Label your graphs with a **T** for the tennis ball and a **C** for the cricket ball.

No calculations are required.

[3 marks]



3

Turn over for the next question

Turn over ►



0 4 . 1

An object is made to rotate by a couple acting about an axis through its centre of mass.

State what is meant by a couple.

[2 marks]

---

---

---

---

---

0 4 . 2

State how a single force applied to an object can cause the object to rotate.

[1 mark]

---

---

---

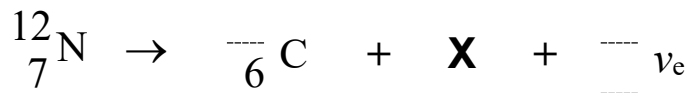
3



05

In the decay of a nucleus of nitrogen-12, an electron neutrino ( $\nu_e$ ) and another particle **X** are emitted from the nucleus.

The decay is shown in the equation below.



05.1

Identify **X**.

[1 mark]

**X** = \_\_\_\_\_

05.2

State the nucleon number of the nucleus produced by this decay.

[1 mark]

nucleon number = \_\_\_\_\_

05.3

Complete below the missing data for the neutrino.

[1 mark]

\_\_\_\_\_  $\nu_e$   
\_\_\_\_\_

05.4

Explain why the existence of the neutrino was suggested before the particle was observed.

[2 marks]

---



---



---



---



---



---



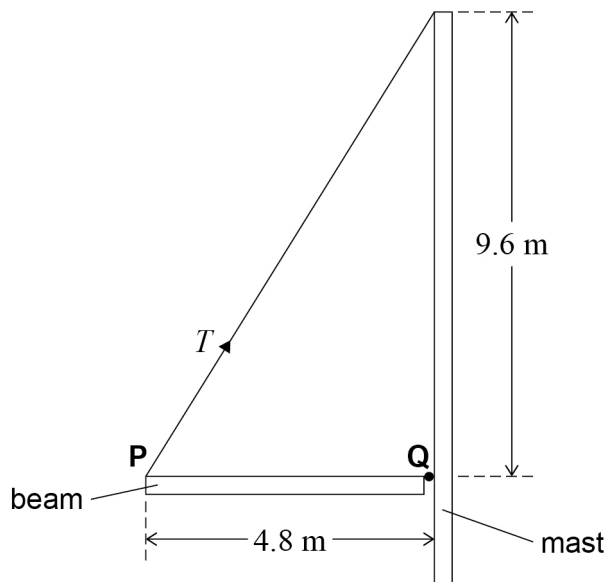
0 6

**Figure 1** shows the mast of a sailing boat.

A uniform horizontal beam **PQ** is attached by a pivot to the mast at **Q**. The beam is supported by a light inextensible rope attached to the top of the mast and to the free end of the beam at **P**.

The tension in the rope is  $T$ .

**Figure 1**



0 6

1

The mass of the beam is 12 kg.

Calculate  $T$  by taking moments about **Q**.

**[4 marks]**

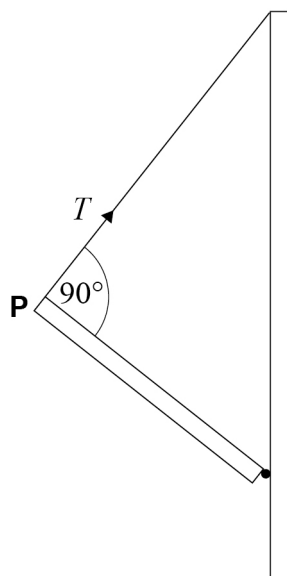
$T =$  \_\_\_\_\_ N



0 6 . 2

The end **P** of the beam is now raised by shortening the rope until it makes an angle of  $90^\circ$  to the beam as shown in **Figure 2**.

**Figure 2**



The magnitude of  $T$  varies as **P** is raised.

Explain the variation in the magnitude of  $T$ .  
Calculations are not required.

**[2 marks]**

---

---

---

---

---

---

---

---



0 7

In an experiment to determine the acceleration due to gravity, a steel ball is dropped from rest.

The time  $t$  taken for the ball to fall a distance  $h$  is measured for a range of values of  $h$ . All of the measurements are shown in **Table 1**.

**Table 1**

| $h / \text{m}$ | $t / \text{s}$ |
|----------------|----------------|
| 0.5            | 0.35           |
| 1.0            | 0.48           |
| 1.5            | 0.6            |
| 2.0            | 0.67           |

0 7 . 1

Describe **three** ways in which the procedure and the recording of results could be improved.

Refer to **Table 1** in your answer.

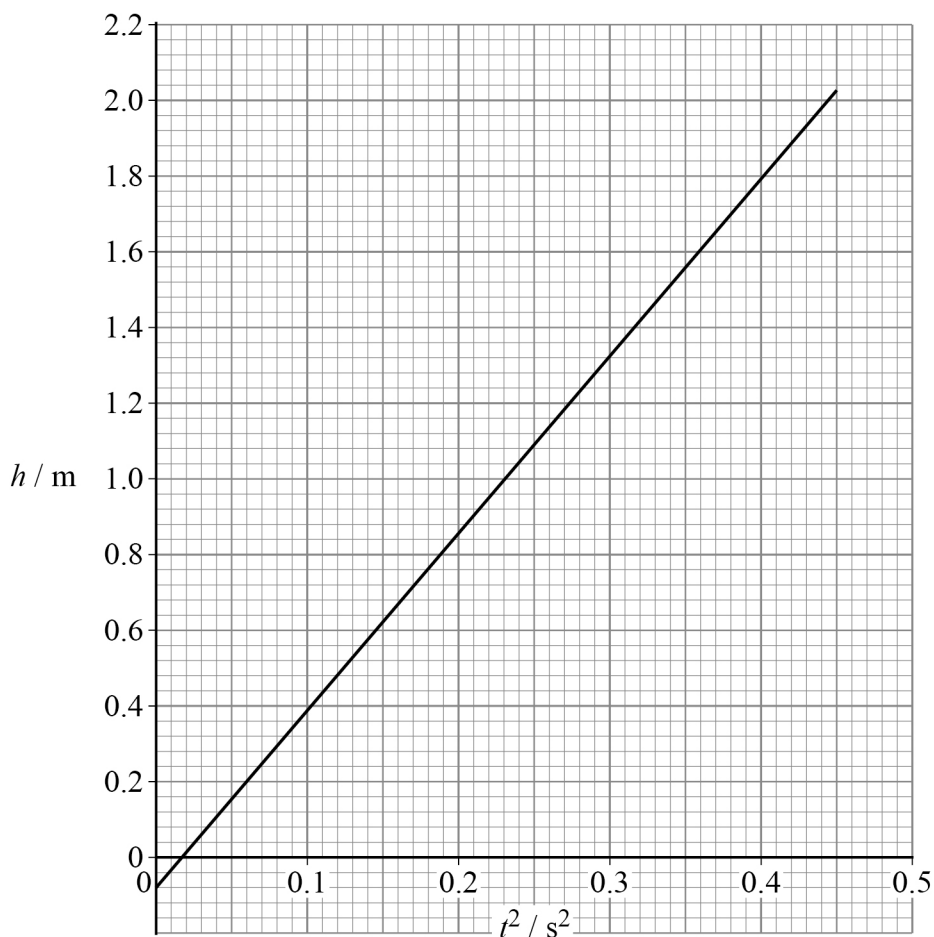
**[3 marks]**

- 1 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_
- 2 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_
- 3 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_



**Figure 3** shows the best-fit line of a graph of  $h$  against  $t^2$  for this experiment.

**Figure 3**



**0 7 . 2** Determine, using **Figure 3**, a value for the acceleration due to gravity.

**[3 marks]**

acceleration due to gravity = \_\_\_\_\_  $\text{m s}^{-2}$

**Question 7 continues on the next page**

**Turn over ►**



**0 7 . 3** The best-fit line in **Figure 3** does not go through the origin.

Suggest a systematic error that could account for this.

**[1 mark]**

---

---

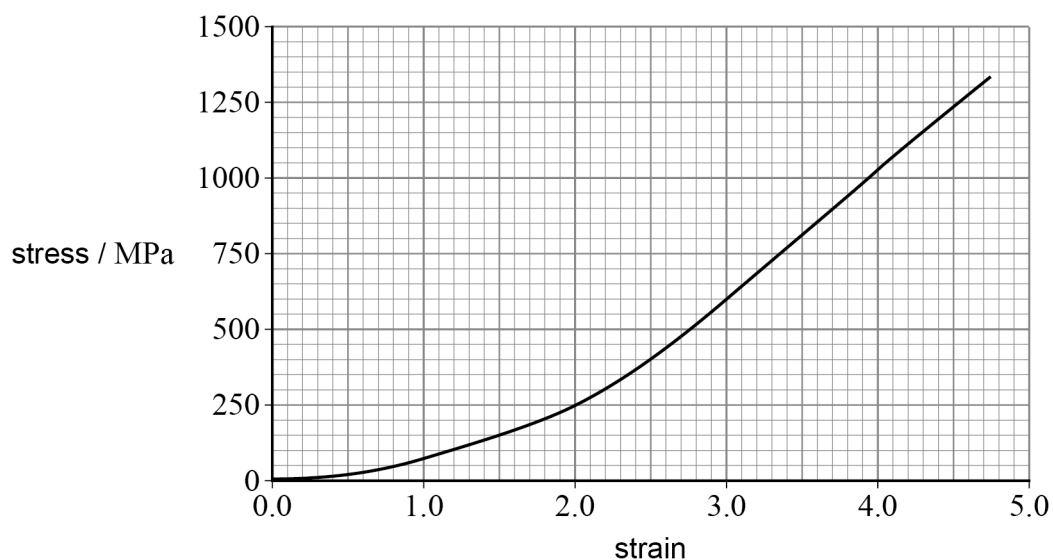
---

**0 7 . 4** Draw a labelled diagram of the arrangement of apparatus you would use to measure  $t$  to an appropriate resolution.

**[2 marks]**

08

A strand of spider web silk is tested by increasing the tension in it until it breaks.  
**Figure 4** shows a stress–strain graph for the strand.

**Figure 4**

08

1

Explain whether or not the graph in **Figure 4** shows that the strand is elastic.

**[2 marks]**


---



---



---



---



---

08

2

The original length of this strand is 1.8 cm.

Show that the length of the strand just before it breaks is approximately 10 cm.

**[2 marks]**

Question 8 continues on the next page

Turn over ►



08.3

The work done per unit volume in stretching the strand is equivalent to the area under a stress–strain curve.

The volume of the strand remains constant as it stretches.

The initial diameter of the strand is  $8.0 \times 10^{-6}$  m.

Determine the energy needed to break the strand.

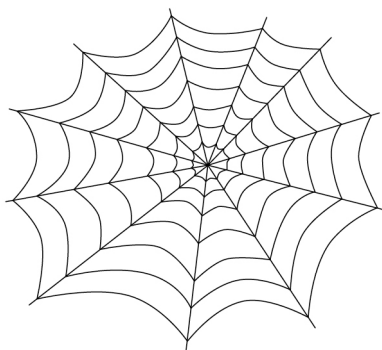
[4 marks]

energy = \_\_\_\_\_ J

08.4

Figure 5 shows the spider's web.

Figure 5



An insect flies into the web. The kinetic energy of the insect is much greater than the value of energy calculated in Question 08.3.

Suggest why the web is unlikely to break.

[1 mark]

---



---

09

A rock is ejected from point **O** on the side of a volcano and hits the sea at **P**.

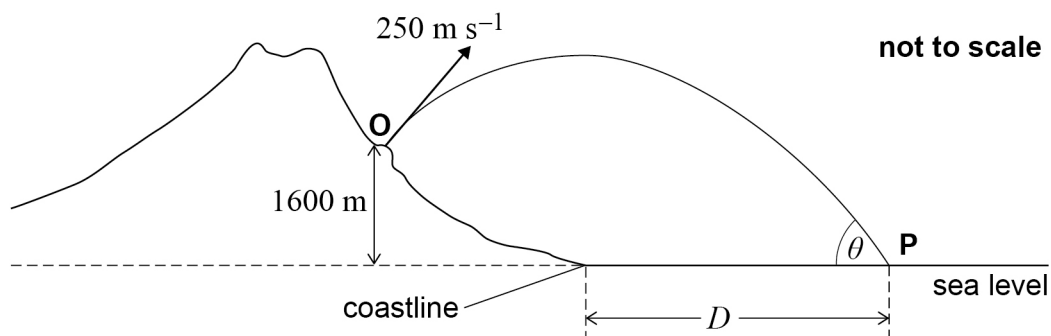
**Figure 6** shows the trajectory of the rock.

The mass of the rock is  $3.70 \text{ kg}$  and its initial speed is  $250 \text{ m s}^{-1}$ .

**O** is  $1600 \text{ m}$  above sea level.

Assume that air resistance is negligible.

**Figure 6**



09.1

Show that the kinetic energy of the rock as it enters the sea at **P** is approximately  $1.7 \times 10^5 \text{ J}$ .

[3 marks]

09.2

Show that the speed of the rock at **P** is approximately  $300 \text{ m s}^{-1}$ .

[1 mark]

Question 9 continues on the next page

Turn over ►



The horizontal component of velocity of the rock at **O** is  $167 \text{ m s}^{-1}$ .

0 9 . 3

Show that the vertical component of the velocity of the rock at **P** is approximately  $260 \text{ m s}^{-1}$ .

[2 marks]

0 9 . 4

The rock enters the sea at **P** at an angle of  $\theta$  to the horizontal.

Calculate  $\theta$ .

[1 mark]

$$\theta = \underline{\hspace{2cm}}^\circ$$

0 9 . 5

The rock reaches its maximum height when it is exactly above the coastline.

Calculate  $D$ , the distance of **P** from the coastline.

[3 marks]

$$D = \underline{\hspace{2cm}} \text{ m}$$

END OF SECTION A



**Section B**

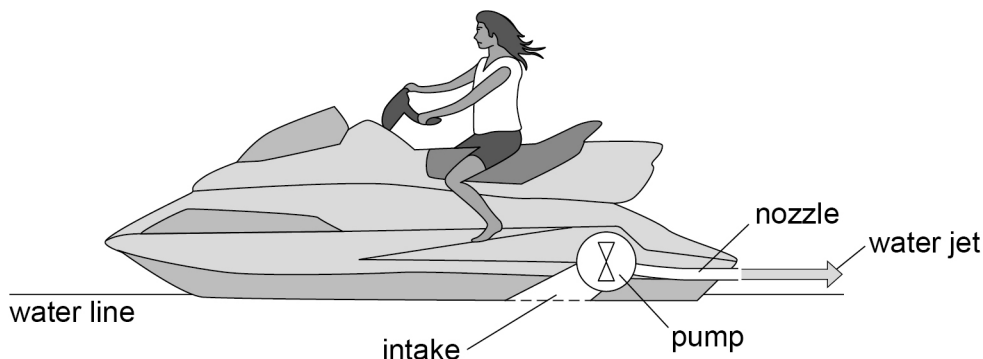
Answer **all** questions in this section.

**1 0**

**Figure 7** shows a jet ski moving horizontally in a straight line. The jet ski takes in water through an intake below the water line. A pump ejects the water horizontally at high speed through a nozzle at the back of the jet ski.

The jet ski and rider have a combined mass of 521 kg.

**Figure 7**

**1 0 . 1**

The rider accelerates the jet ski uniformly from rest to its top speed of  $27 \text{ m s}^{-1}$  in a distance of 63 m.

Show that the resultant force accelerating the jet ski is approximately 3.0 kN.

**[3 marks]**

Question 10 continues on the next page

Turn over ►

**1 0 . 2** Explain, in terms of momentum, how the jet ski is propelled forwards.

**[2 marks]**

---

---

---

---

---

**1 0 . 3** During the acceleration, the jet ski ejects 78 kg of water per second through the nozzle.

Calculate the speed at which the water is ejected from the nozzle.

**[2 marks]**

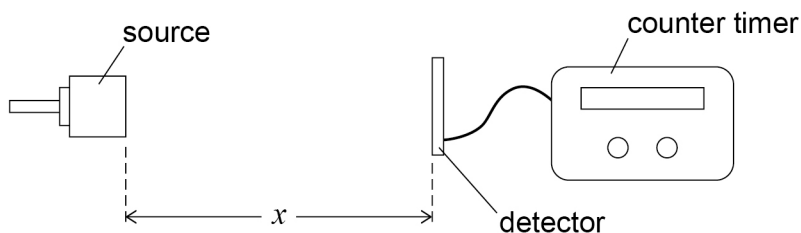
speed = \_\_\_\_\_ m s<sup>-1</sup>

      
**7**



1 1

**Figure 8** shows the apparatus used to verify the inverse-square law for gamma rays. The distance  $x$  between the source and a detector is measured. The detector is connected to a counter timer and the count rate  $C$  is measured for different values of  $x$ .

**Figure 8**

A student suggests that  $C$  is given by  $C = \frac{k}{x^2}$  where  $k$  is a constant.

1 1

. 1

The count rate was measured three times for  $x = 0.520$  m.

**Table 2** shows the data recorded.

**Table 2**

|                              |      |      |       |
|------------------------------|------|------|-------|
| <b>Elapsed time / minute</b> | 5.00 | 5.00 | 10.00 |
| <b>Total count</b>           | 2246 | 2263 | 4458  |

Calculate the mean count rate in counts per minute.

**[1 mark]**

mean count rate = \_\_\_\_\_ counts per minute

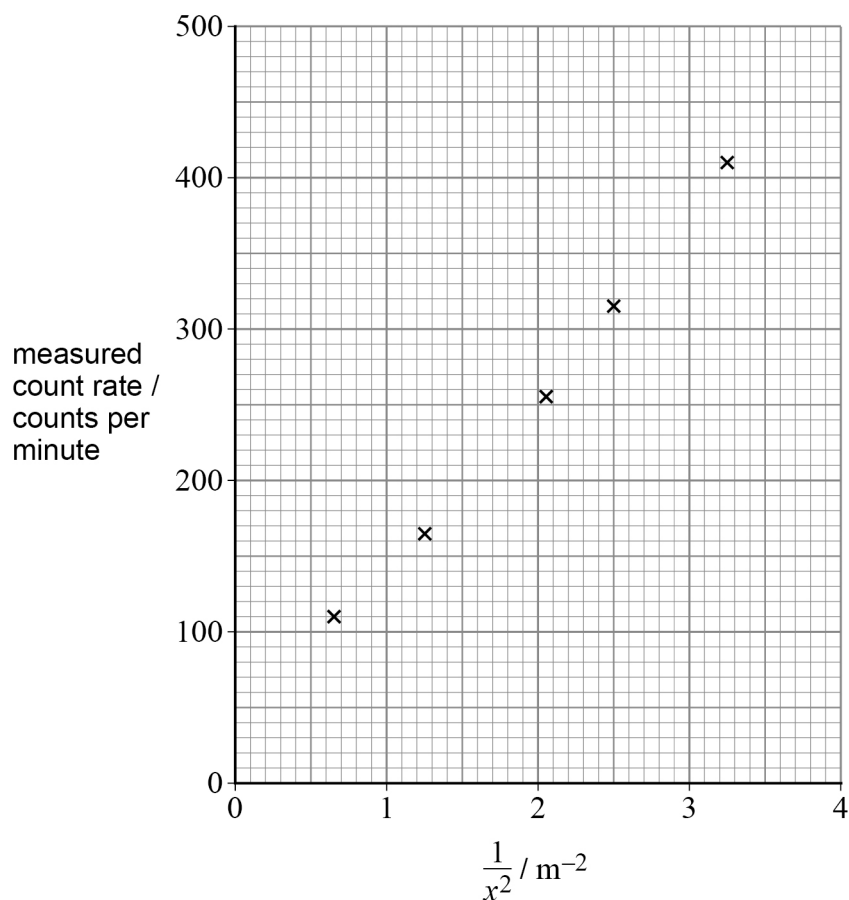
**Question 11 continues on the next page**

**Turn over ►**



**Figure 9** shows a plot of  $C$  against  $\frac{1}{x^2}$  for all of the other measured values of  $C$  and  $x$ .

**Figure 9**



**1 1 . 2** Plot, on **Figure 9**, your answer to Question **11.1**.

[1 mark]

**1 1 . 3** Draw a line of best fit on **Figure 9**.

[1 mark]



**1 1 . 4** Determine the gradient of your best-fit line on **Figure 9**.

**[3 marks]**

gradient = \_\_\_\_\_ counts min<sup>-1</sup> m<sup>2</sup>

**1 1 . 5** Explain whether or not **Figure 9** supports the student's suggestion that  $C = \frac{k}{x^2}$ .

**[1 mark]**

---



---



---

**1 1 . 6** State the physical quantity represented by the intercept on the  $y$ -axis.

**[1 mark]**

---



---

**1 1 . 7** Determine the value of the intercept on the  $y$ -axis.

**[1 mark]**

value of the intercept = \_\_\_\_\_ counts per minute

**END OF SECTION B**



## Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional pages for this working.

- 1 2** A 3 N force and a 4 N force act on an object at right angles to each other. A third force of 8 N also acts on the object.  
All of the forces are coplanar.

What is a possible resultant force on the object?

[1 mark]

**A** 1 N ☐

**B** 10 N ☐

**C** 14 N ☐

**D** 15 N ☐

- 1 3** What is **not** a possible unit for kinetic energy?

[1 mark]

**A**  $\mu\text{J}$  ☐

**B**  $\text{kg m s}^{-2}$  ☐

**C** eV ☐

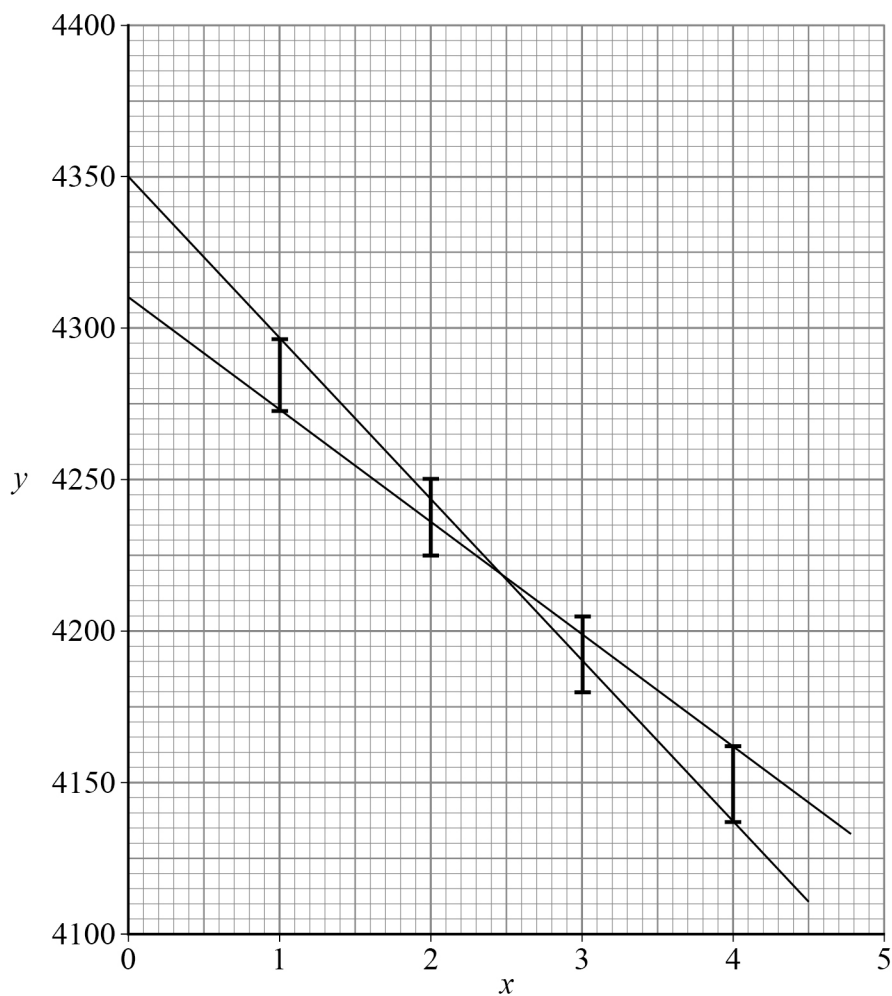
**D**  $\text{Ws}$  ☐



1 4

What is the percentage uncertainty in the value of the  $y$ -intercept that can be deduced from this graph?

[1 mark]



**A** 0.5%

**B** 0.9%

**C** 5%

**D** 9%



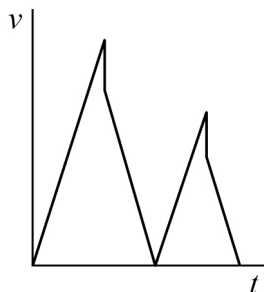
1 5

A ball is dropped onto a horizontal floor.  
The ball bounces twice. The ball loses some energy each time it bounces.  
Ignore the effect of air resistance and the time that the ball is in contact with the floor.

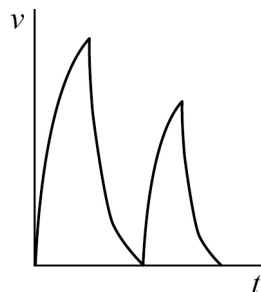
Which graph shows the variation of velocity  $v$  of the ball with time  $t$ ?

[1 mark]

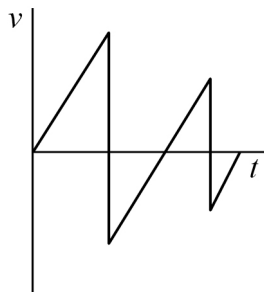
**A**



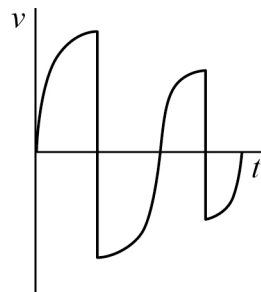
**B**



**C**



**D**



**A**

☐

**B**

☐

**C**

☐

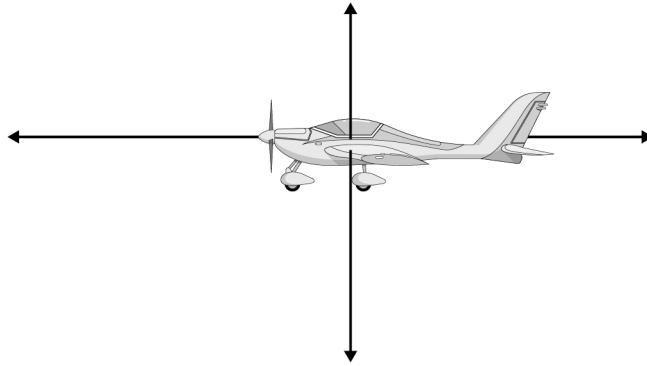
**D**

☐


1 6

The diagram shows an aircraft and the forces acting on it.

The lengths of the arrows indicate the magnitude of the forces.



Which row describes the horizontal and vertical components of velocity of the aircraft?

[1 mark]

|          | Horizontal component<br>of velocity | Vertical component<br>of velocity |                       |
|----------|-------------------------------------|-----------------------------------|-----------------------|
| <b>A</b> | constant to the left                | constant downwards                | <input type="radio"/> |
| <b>B</b> | constant to the left                | increasing downwards              | <input type="radio"/> |
| <b>C</b> | increasing to the left              | constant downwards                | <input type="radio"/> |
| <b>D</b> | increasing to the left              | increasing downwards              | <input type="radio"/> |

Turn over for the next question

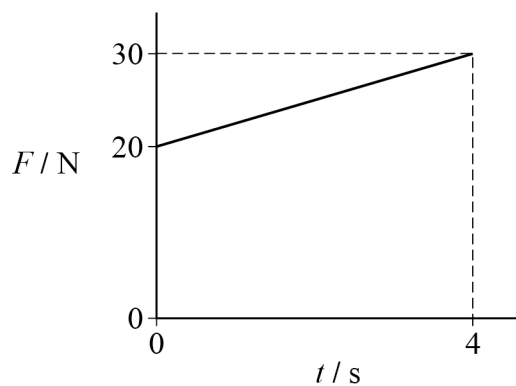
Turn over ►



**1 7**

A force  $F$  acts on an object.

The graph shows the variation of  $F$  with time  $t$ .



What is the impulse received by the object during the time shown by the graph?

**[1 mark]**

**A**  $2.5 \text{ kg m s}^{-1}$

☐

**B**  $10 \text{ kg m s}^{-1}$

☐

**C**  $30 \text{ kg m s}^{-1}$

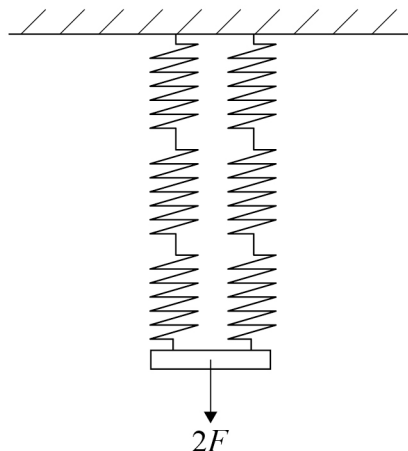
☐

**D**  $100 \text{ kg m s}^{-1}$

☐

1 8

When a single spring of stiffness  $k$  is stretched by a force  $F$ , the extension is  $\Delta l$ .  
The diagram shows a network of six single springs of stiffness  $k$  being stretched by a force  $2F$ .



What is the total extension of the network of springs?

[1 mark]

**A**  $\frac{4}{3} \Delta l$

☐

**B**  $\frac{3}{2} \Delta l$

☐

**C**  $3 \Delta l$

☐

**D**  $6 \Delta l$

☐

Turn over for the next question

Turn over ►



- 1 9** When an elevator of mass 400 kg raises a load of mass 900 kg the efficiency is 60%. The useful work done by the elevator is the work done in lifting the load. When moving through a height  $h$ , the total energy input to the elevator is  $3.4 \times 10^5$  J.

What is  $h$ ?

[1 mark]

**A** 16 m

☐

**B** 23 m

☐

**C** 44 m

☐

**D** 64 m

☐

- 2 0** The table shows observations and deductions from the Rutherford scattering experiment when carried out with copper-63 and gold-197 atoms.

Which row has a correct observation and a deduction that may be made from that observation?

[1 mark]

|          | Observation                                                                           | Deduction                                                   |                       |
|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------|
| <b>A</b> | a small number of alpha particles are deviated through angles of more than $90^\circ$ | atomic nuclei are very small                                | <input type="radio"/> |
| <b>B</b> | many alpha particles are deviated through angles of more than $90^\circ$              | atomic nuclei carry a positive charge                       | <input type="radio"/> |
| <b>C</b> | copper foil deflects particles through greater angles than gold foil                  | a copper nucleus contains more nucleons than a gold nucleus | <input type="radio"/> |
| <b>D</b> | a minority of alpha particles pass through the foil without deviation                 | atomic nuclei are small and positively charged              | <input type="radio"/> |



**2 1** Which pair of particles has the same magnitude of specific charge?

**[1 mark]**

**A**  ${}^3_2\text{He}$  nucleus and  ${}^3_1\text{H}$  nucleus

☐

**B**  ${}^2_1\text{H}$  nucleus and  ${}^1_1\text{H}$  nucleus

☐

**C** proton and positron

☐

**D** beta particle and positron

☐

**2 2** Which row shows two particles that can mutually annihilate and the energy produced in their annihilation?

**[1 mark]**

|          | 1st particle | 2nd particle | Energy produced / MeV |                          |
|----------|--------------|--------------|-----------------------|--------------------------|
| <b>A</b> | proton       | proton       | 1860                  | <input type="checkbox"/> |
| <b>B</b> | proton       | proton       | 930                   | <input type="checkbox"/> |
| <b>C</b> | proton       | antiproton   | 1860                  | <input type="checkbox"/> |
| <b>D</b> | proton       | antiproton   | 930                   | <input type="checkbox"/> |

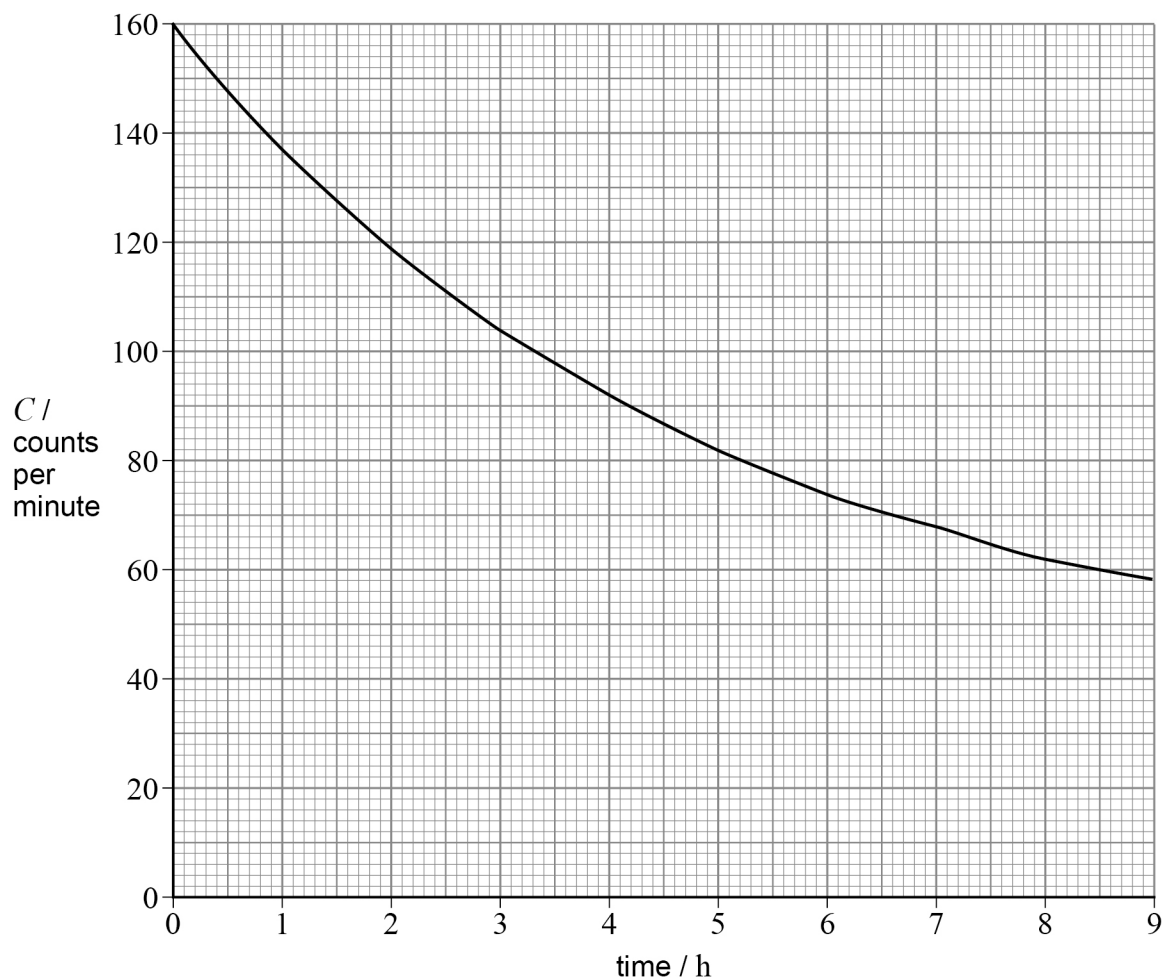
**Turn over for the next question**

**Turn over ►**



**2 3**

A detector is placed near a source of radiation.  
The total count rate  $C$  is plotted against time.  
The background count rate is 40 counts per minute.



What is the half-life of the source?

[1 mark]

**A** 1.9 h

☐

**B** 3.3 h

☐

**C** 5.2 h

☐

**D** 8.8 h

☐

**2 4**

A detector is placed near to a radioactive source **S**.

Absorbers are placed between **S** and the detector.

The table shows the corrected count rate for radiation from **S** with and without absorbers present.

| Absorber               | Corrected count rate / counts s <sup>-1</sup> |
|------------------------|-----------------------------------------------|
| none                   | 480                                           |
| one sheet of thin card | 340                                           |
| 4 mm of aluminium      | 340                                           |

What is emitted from **S**?

[1 mark]

**A**  $\alpha$  radiation only

☐

**B**  $\alpha$  and  $\beta$  radiations only

☐

**C**  $\alpha$  and  $\gamma$  radiations only

☐

**D**  $\alpha$ ,  $\beta$  and  $\gamma$  radiations

☐**2 5**

Which statement concerning background radiation is correct?

[1 mark]

**A** Background radiation is made up of alpha, beta and gamma radiation only.

☐

**B** All background radiation comes from naturally occurring sources.

☐

**C** Some background radiation comes from sources not on the Earth.

☐

**D** All background radiation can be absorbed by a lead screen.

☐

[illegible]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

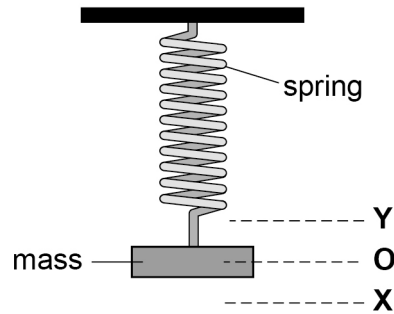
Copyright © 2021 Oxford International AQA Examinations and its licensors. All rights reserved.



2 1 1 X P H 0

**Figure 1** shows a mass hanging in its equilibrium position **O** on a light, vertical spring.

**Figure 1**



The mass is pulled down to position **X** and then released. The mass reaches a maximum height at position **Y**.

Describe the energy transfers in the system as the mass moves from **X** to **Y**.

**[4 marks]**

---

---

---

---

---

---

---

---

---

---

0 2 . 1

State the condition that is necessary for momentum to be conserved during a collision.

[1 mark]

---

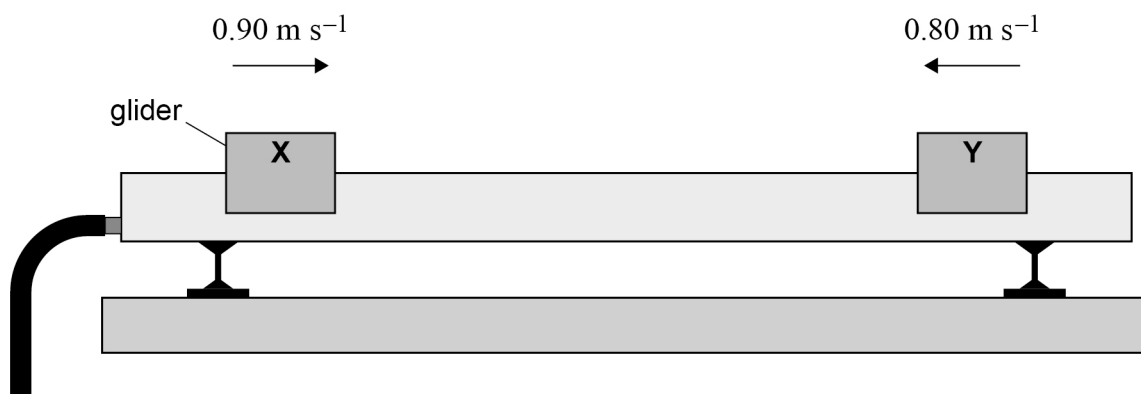
---

0 2 . 2

**Figure 2** shows two gliders **X** and **Y** moving horizontally on a frictionless track. Initially:

- **X** is moving to the right at a speed of  $0.90 \text{ m s}^{-1}$
- **Y** is moving to the left at a speed of  $0.80 \text{ m s}^{-1}$ .

**Figure 2**



The gliders collide.

After the collision:

- **X** moves to the left at a speed of  $0.30 \text{ m s}^{-1}$
- **Y** moves to the right at a speed of  $0.15 \text{ m s}^{-1}$ .

The mass of **X** is  $0.050 \text{ kg}$ .

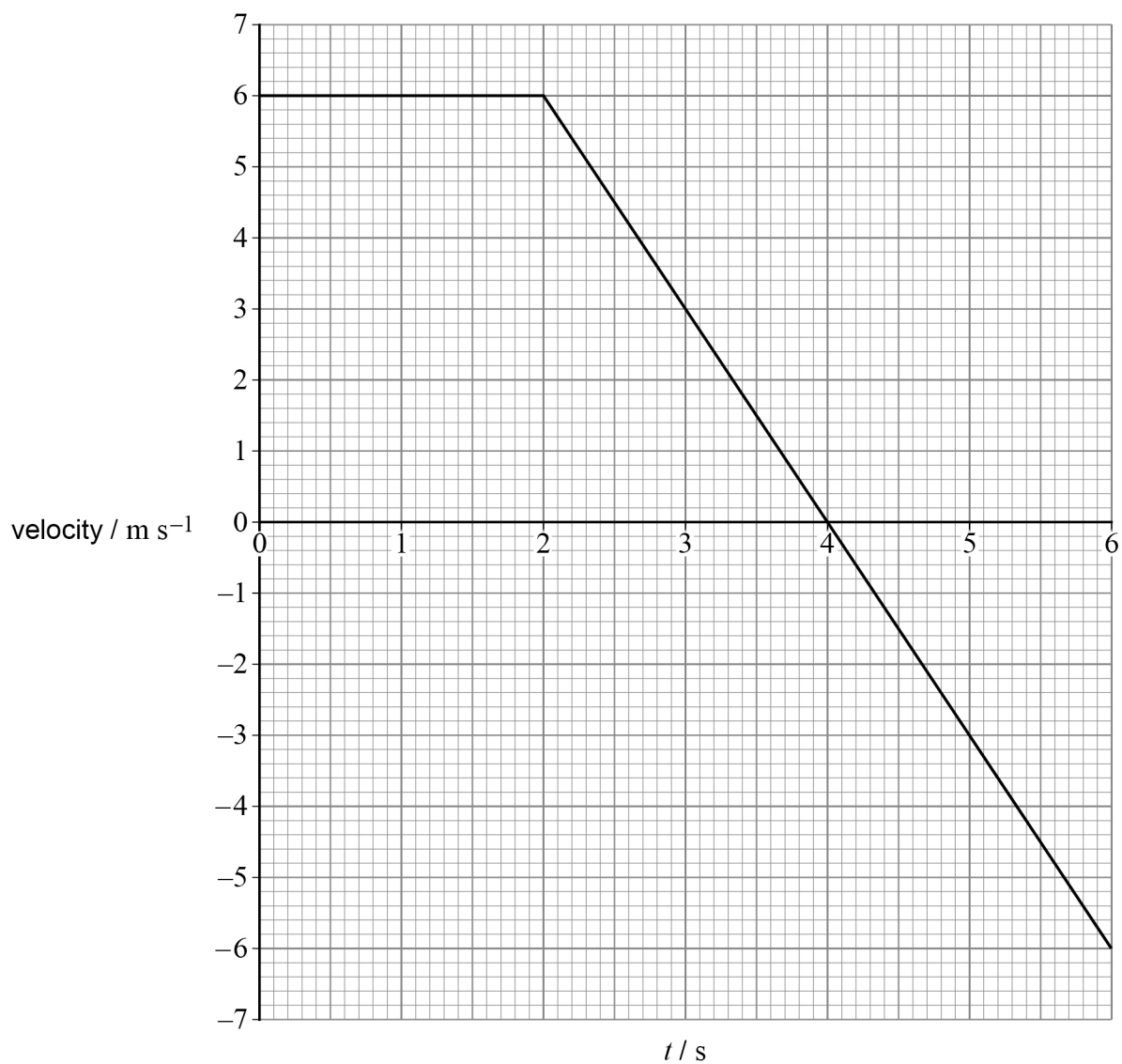
Calculate the mass of **Y**.

[3 marks]

mass of **Y** = \_\_\_\_\_ kg

**Figure 3** is a velocity–time graph for a moving object. The displacement of the object is zero at time  $t = 0$

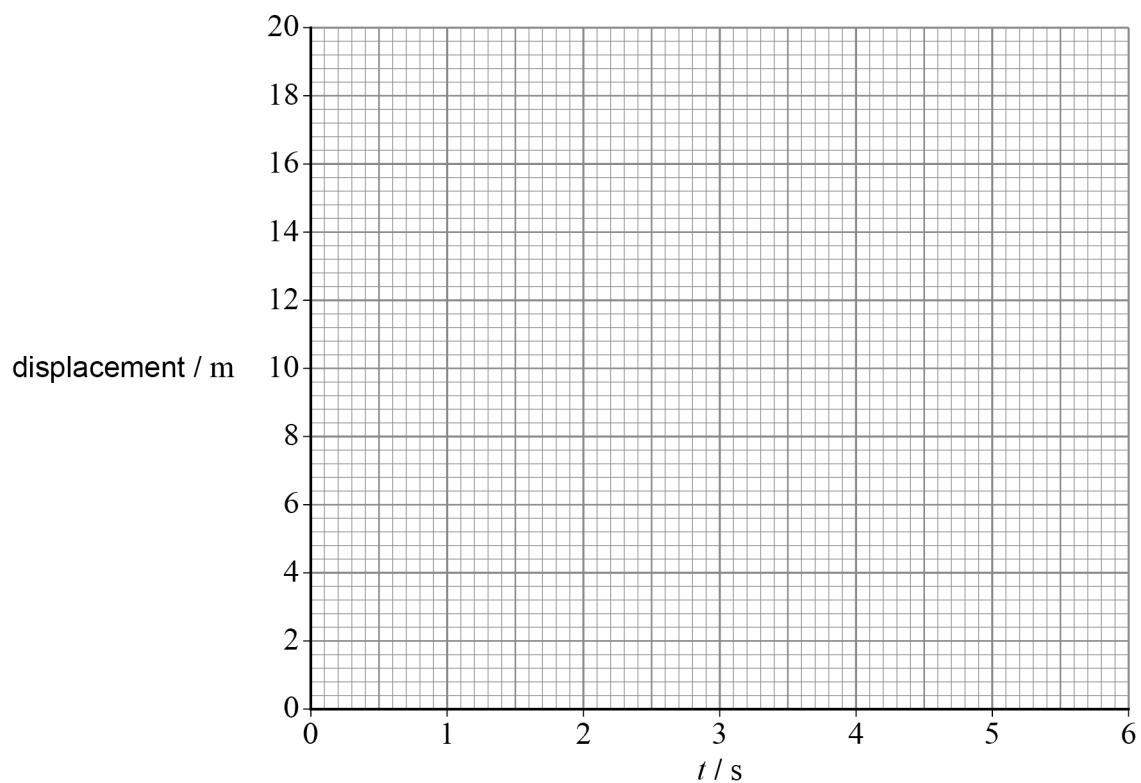
**Figure 3**



Plot, on **Figure 4**, the displacement–time graph for the object between 0 and 6 s.

**[4 marks]**

**Figure 4**



      
**4**

**Turn over for the next question**

**Turn over ►**



**0 4 . 1** State the products of the decay of a free neutron.

**[2 marks]**

Two different experimental methods **A** and **B** are used to determine the average time  $t_n$  for the decay of a free neutron. **Table 1** shows the result of each method.

**Table 1**

| Method   | $t_n / \text{s}$ |
|----------|------------------|
| <b>A</b> | $887.7 \pm 2.2$  |
| <b>B</b> | $878.5 \pm 0.8$  |

**0 4 . 2** Compare the precision of the measurement of  $t_n$  determined by methods **A** and **B**.

**[1 mark]**

**0 4 . 3** State what is meant by a reproducible measurement.

**[1 mark]**



0 4 . 4

Discuss whether methods **A** and **B** have provided a reproducible measurement of  $t_n$ .

**[2 marks]**

---

---

---

---

---

---

6

**Turn over for the next question**

**Turn over ►**



0 5

Gold ions collide in a particle accelerator. Each ion is made by removing 77 electrons from an atom of gold ( $^{197}_{79}\text{Au}$ ) in two stages.

0 5 . 1

In the first stage, ions of  $^{197}_{79}\text{Au}^{32+}$  are made.

Calculate the number of electrons in an  $^{197}_{79}\text{Au}^{32+}$  ion.

[1 mark]

number of electrons = \_\_\_\_\_

0 5 . 2

In the second stage, more electrons are removed to produce ions of  $^{197}_{79}\text{Au}^{77+}$ .

Calculate the specific charge of an  $^{197}_{79}\text{Au}^{77+}$  ion.

[3 marks]

specific charge = \_\_\_\_\_  $\text{C kg}^{-1}$



Collisions between the gold ions produce nuclei of antimatter helium-4  $\left({}_2^4\text{He}\right)$ .

0 5 . 3

State the constituents of one nucleus of antimatter helium-4

[2 marks]

---

---

---

---

0 5 . 4

Collisions between the gold ions lead to two annihilations:

- a nucleus of antimatter helium with a nucleus of helium
- an antiproton with a proton.

Both annihilations produce radiation.

The kinetic energy of the annihilating particles is negligible.

Compare the radiation produced as a result of the two different annihilations.

[3 marks]

---

---

---

---

---

---

---

---

---

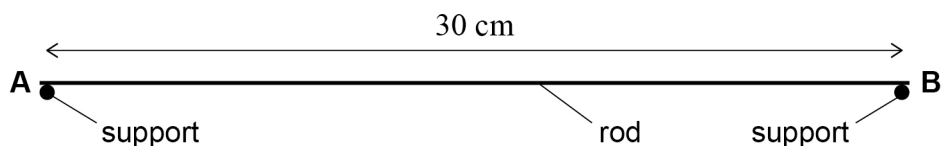
---



0 6

**Figure 5** shows a uniform rod **AB** of length 30 cm and mass 6.0 g.  
The rod is supported at each end.

**Figure 5**



0 6

1

Calculate the magnitude of the force of **one** support on the rod.

[1 mark]

magnitude of force = \_\_\_\_\_ N

0 6

2

Explain whether the forces from the supports on the rod produce a couple.

[2 marks]

---

---

---

---

0 6

3

State the principle of moments.

[2 marks]

---

---

---

---

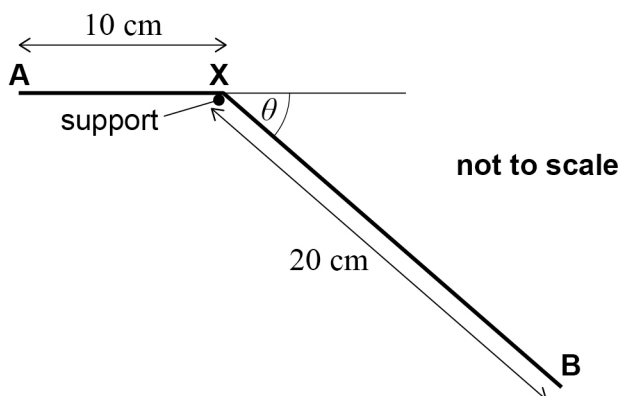


The rod is now bent through an angle  $\theta$  at **X** as shown in **Figure 6**.  
When the rod is balanced on a single support at **X**, section **AX** is horizontal.

length of **AX** = 10 cm

length of **XB** = 20 cm

**Figure 6**



Calculate  $\theta$ .

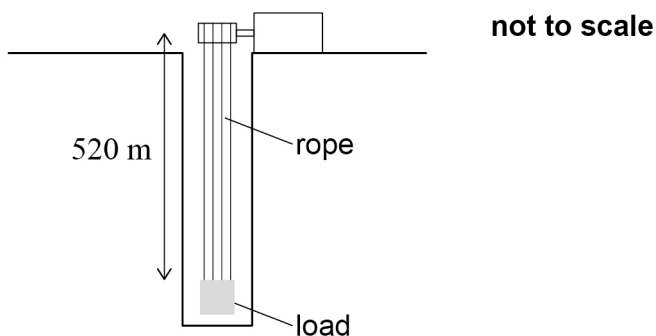
[3 marks]

$\theta =$  \_\_\_\_\_ degrees

0 7

A system designed to store energy uses a large load suspended from four steel ropes in an underground vertical tunnel. Energy is stored when the load is lifted.

**Figure 7**

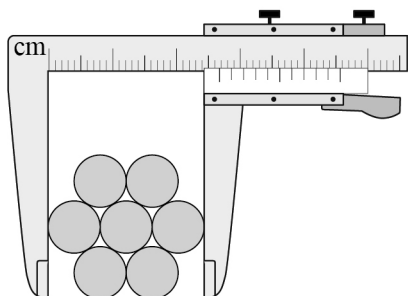


0 7 . 1

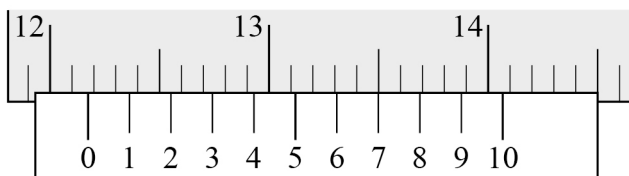
**Figure 8** shows a vernier caliper measuring the maximum width of one steel rope. The rope is made from seven identical wires.

**Figure 9** shows a larger view of the scale of the caliper.

**Figure 8**



**Figure 9**



Show, using **Figure 9**, that the cross-sectional area of this rope is approximately  $9.0 \times 10^{-3} \text{ m}^2$ .

**[3 marks]**



0 7 . 2

**Figure 7** shows the load at its lowest point. The vertical length of each steel rope is 520 m.

Show that the total mass of the four ropes is approximately  $9 \times 10^4$  kg.

$$\text{density of steel} = 4.7 \times 10^3 \text{ kg m}^{-3}$$

**[2 marks]**

The maximum tension in each rope is  $1.4 \times 10^6$  N when the load is accelerated from its lowest point.

The mass of the load is  $4.5 \times 10^5$  kg.

0 7 . 3

Calculate the acceleration of the load.

**[4 marks]**

acceleration = \_\_\_\_\_  $\text{m s}^{-2}$

**Question 7 continues on the next page**

**Turn over ►**



0 7 . 4

The breaking stress of the steel used to make the four ropes is 890 MPa.  
For safety reasons, the maximum stress in each rope during the acceleration must be less than  $\frac{1}{5}$  of the breaking stress.

Deduce whether these ropes are safe.

[2 marks]

---

---

---

0 7 . 5

It is suggested that the load and ropes will store 4 MW h of energy when the load has been lifted by 520 m.

Deduce whether this suggestion is correct.

[4 marks]

---

---

---

END OF SECTION A

## Section B

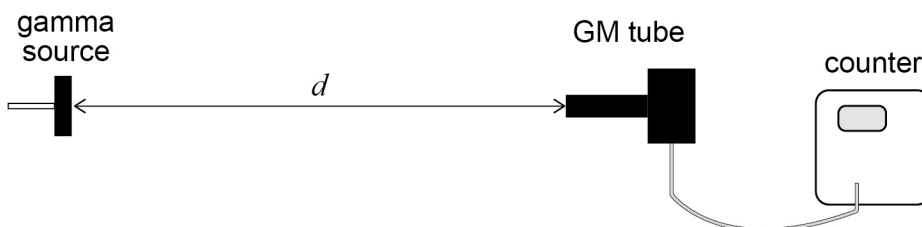
Answer **all** questions in this section.

0 8

**Figure 10** shows an arrangement used to investigate the inverse-square law for gamma radiation.

The count rate is measured for a range of distances  $d$  between the source and the Geiger Müller (GM) tube.

Figure 10



0 8 . 1

A background count is determined at the start and at the end of the investigation.

Explain why.

[1 mark]

---



---



---

When  $d = 9.0$  cm the measured count rate is  $84.49 \text{ s}^{-1}$  and the corrected count rate  $C_R$  is  $84.16 \text{ s}^{-1}$ .

0 8 . 2

Deduce the background count measured over a time period of 5 minutes.

[1 mark]

background count = \_\_\_\_\_

**Question 8 continues on the next page**

**Turn over ►**



**Table 2** shows values of  $d$ ,  $C_R$  and  $\frac{1}{\sqrt{C_R}}$ .

**Table 2**

| $d / \text{cm}$ | $C_R / \text{s}^{-1}$ | $\frac{1}{\sqrt{C_R}} / \text{s}^{\frac{1}{2}}$ |
|-----------------|-----------------------|-------------------------------------------------|
| 9.0             | 84.16                 | 0.109                                           |
| 12.0            | 56.93                 | 0.133                                           |
| 16.0            | 35.19                 | 0.169                                           |
| 20.0            | 23.98                 |                                                 |
| 24.0            | 18.39                 |                                                 |
| 28.0            | 14.79                 |                                                 |

0 8 . 3

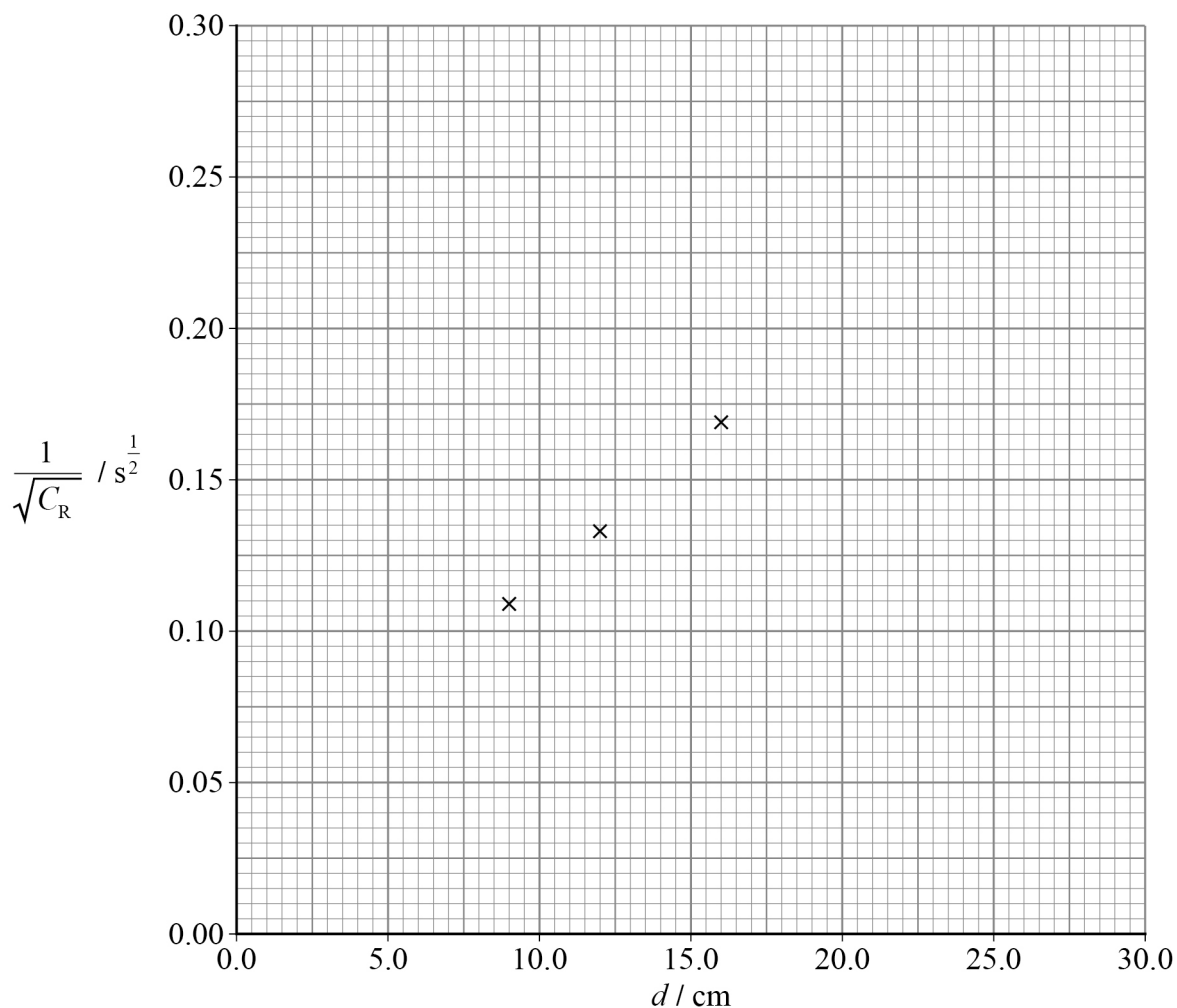
Complete **Table 2**.

[1 mark]



**Figure 11** is a plot of  $\frac{1}{\sqrt{C_R}}$  against  $d$ .

**Figure 11**



**0 8 . 4** Plot on **Figure 11** the last three points from your completed **Table 2**.

[1 mark]

**0 8 . 5** Draw on **Figure 11** the line of best fit.

[1 mark]

**Question 8 continues on the next page**



0 8 . 6

Discuss whether your graph supports the inverse-square law.

[2 marks]

---

---

---

---

---

---

---

7



**Turn over for the next question**

*Do not write  
outside the  
box*

**DO NOT WRITE ON THIS PAGE  
ANSWER IN THE SPACES PROVIDED**

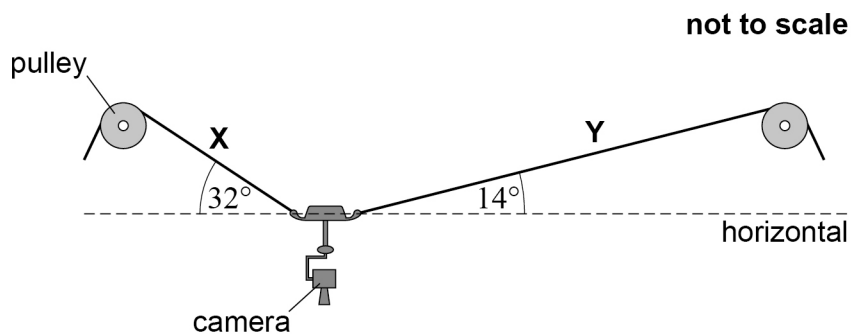
**Turn over ►**



0 9

**Figure 12** shows a camera above a sports field. The camera is supported by two cables **X** and **Y** that pass over smooth pulleys. **X** and **Y** have negligible mass.

**Figure 12**



The camera is held stationary.

The tension in **X** is 450 N and **X** makes an angle of  $32^\circ$  to the horizontal.

**Y** makes an angle of  $14^\circ$  to the horizontal.

0 9 . 1

Show that the tension in **Y** is approximately 400 N.

**[2 marks]**



0 9 . 2

Determine, using a scale diagram, the weight of the camera.

[3 marks]

weight = \_\_\_\_\_ N

Question 9 continues on the next page

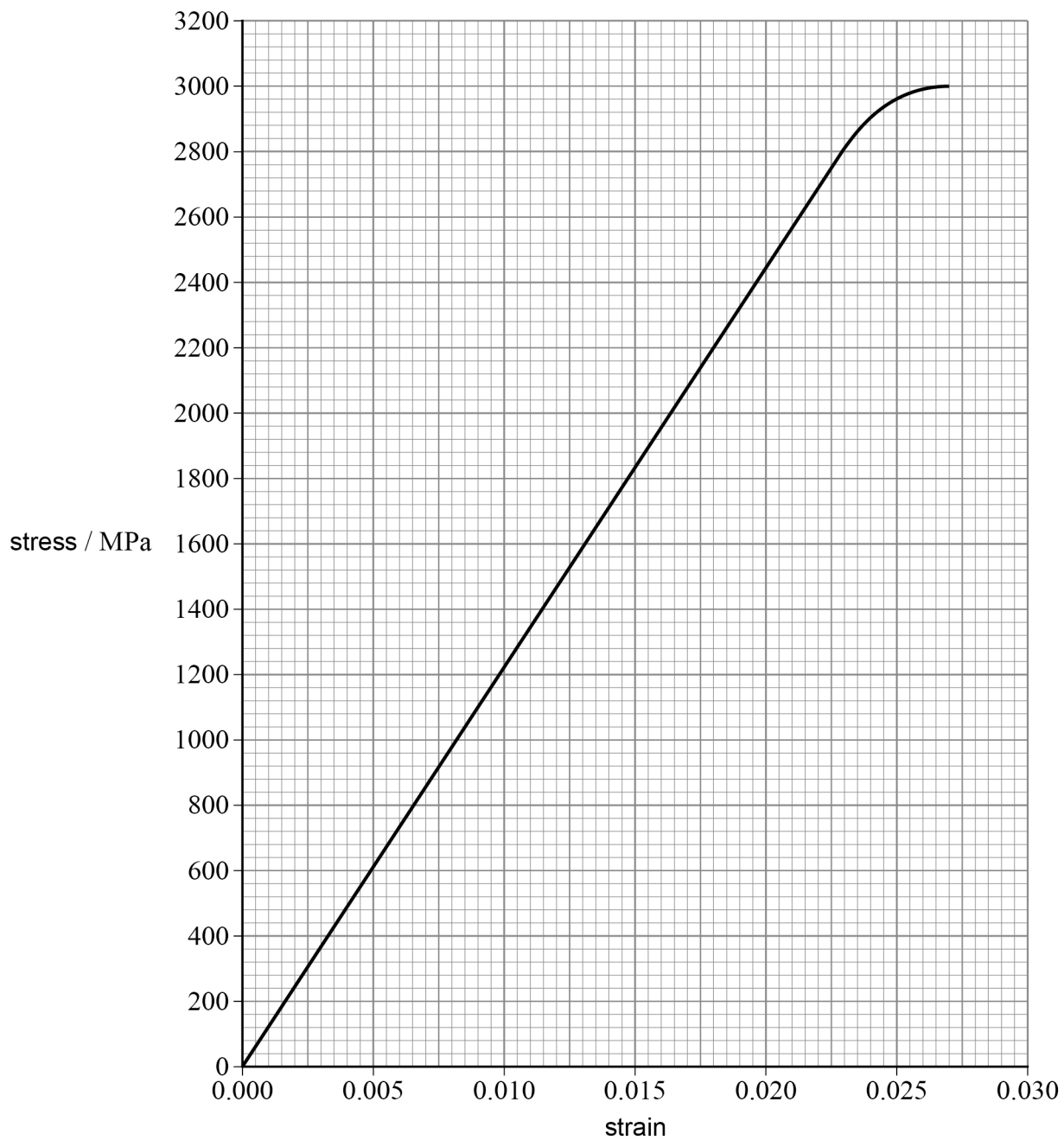
Turn over ►



09.3

Figure 13 shows the variation of stress with strain for the material used to make X.

Figure 13



**X** has a cross-sectional area of  $6.7 \times 10^{-6} \text{ m}^2$  and an unstretched length of 75 m.  
The tension in **X** is 450 N.

Determine the extension of **X**.

[3 marks]

extension of **X** = \_\_\_\_\_ m

0 9 . 4

The camera is moved horizontally to the right to a new stationary position.

Describe the change in the tension in **X** and the change in the tension in **Y**.

[1 mark]

---



---



---

9

END OF SECTION B

Turn over ►



## Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

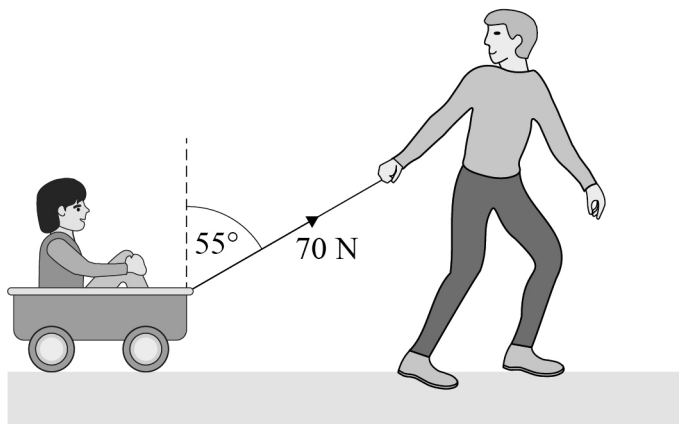


If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional pages for this working.

- 1 0** A father pulls his child in a trolley at a constant speed of  $2.0 \text{ m s}^{-1}$  across a flat surface.  
The father exerts a force of  $70 \text{ N}$  on the handle of the trolley for 2 minutes.  
The handle of the trolley is at an angle of  $55^\circ$  to the vertical.



What is the work done by the father against the resistive forces that act on the trolley?

[1 mark]

**A**  $9.6 \text{ kJ}$

☐

**B**  $14 \text{ kJ}$

☐

**C**  $17 \text{ kJ}$

☐

**D**  $21 \text{ kJ}$

☐


**1 1**

Which row shows two scalar quantities and one vector quantity?

**[1 mark]**

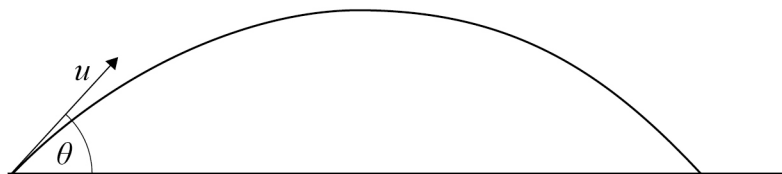
|          |              |          |              |                       |
|----------|--------------|----------|--------------|-----------------------|
| <b>A</b> | momentum     | mass     | velocity     | <input type="radio"/> |
| <b>B</b> | acceleration | speed    | time         | <input type="radio"/> |
| <b>C</b> | work done    | force    | displacement | <input type="radio"/> |
| <b>D</b> | power        | velocity | force        | <input type="radio"/> |

**1 2**An object with mass  $m$  falls through the air with a terminal speed  $v$ .How much energy is transferred to the air in time  $t$ ?**[1 mark]****A**  $mgt$  ☐**B**  $mgvt$  ☐**C**  $\frac{mv}{t}$  ☐**D**  $mvt$  ☐**1 3**A motor is supplied with 5.2 kJ of energy in 2.5 minutes.  
The efficiency of the motor is 35%.

What is the useful power output of the motor?

**[1 mark]****A** 12 W ☐**B** 23 W ☐**C** 73 W ☐**D** 99 W ☐

- 1 4** An object is launched at an initial velocity of  $u$  at an angle of  $\theta$  to the horizontal.



The initial vertical displacement of the object is zero.

At time  $T$ , the vertical displacement of the object becomes zero again.

The effect of air resistance is negligible.

What is the vertical component of the velocity of the object at time  $T$ ?

[1 mark]

**A**  $-u\sin\theta$  ☐

**B**  $-u\cos\theta$  ☐

**C**  $u\sin\theta$  ☐

**D**  $u\cos\theta$  ☐

- 1 5** An unpowered rollercoaster car is released from rest and descends through a height of 55 m.

What is the maximum possible speed of the car?

[1 mark]

**A**  $11 \text{ m s}^{-1}$  ☐

**B**  $16 \text{ m s}^{-1}$  ☐

**C**  $23 \text{ m s}^{-1}$  ☐

**D**  $33 \text{ m s}^{-1}$  ☐

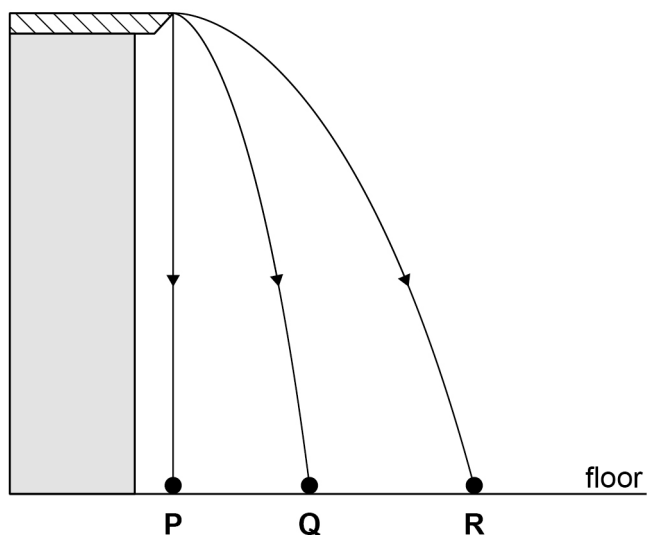


**1 6** Three identical objects **P**, **Q** and **R** leave the edge of a table at the same time.

**P** is released from rest and falls vertically.

**Q** and **R** are projected horizontally at different speeds.

The objects move in parallel vertical planes.



Which statement is correct?

[1 mark]

**A** **P** hits the floor before **Q** and **R**.

☐

**B** **Q** hits the floor before **R**.

☐

**C** **Q** hits the floor with the same speed as **R**.

☐

**D** **R** hits the floor with the greatest speed.

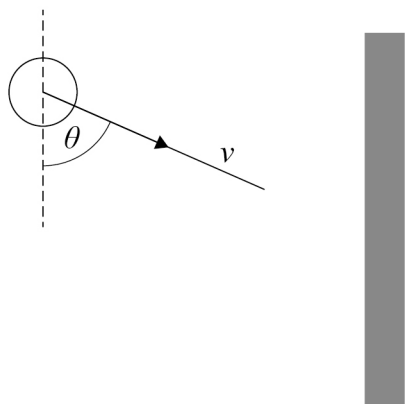
☐

Turn over for the next question

Turn over ►



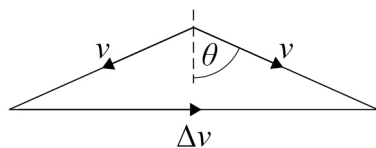
**1 7** An atom travelling at speed  $v$  collides elastically with the wall of a container.



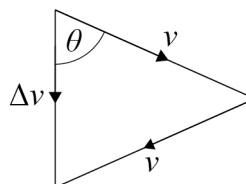
Which diagram shows the change in velocity  $\Delta v$  of the atom for the collision?

[1 mark]

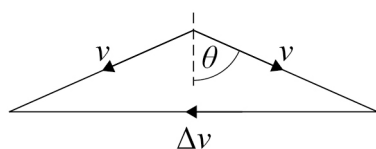
**A**



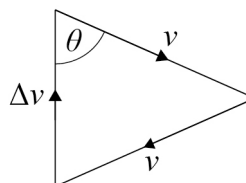
**B**



**C**



**D**



**A**

☐

**B**

☐

**C**

☐

**D**

☐


1 8

Three coplanar forces  $F_1$ ,  $F_2$  and  $F_3$  act on a point object.

Which combination of these forces **cannot** produce a resultant force of zero?

[1 mark]

|          | $F_1$ | $F_2$ | $F_3$ |                       |
|----------|-------|-------|-------|-----------------------|
| <b>A</b> | 1     | 10    | 10    | <input type="radio"/> |
| <b>B</b> | 6     | 8     | 10    | <input type="radio"/> |
| <b>C</b> | 10    | 10    | 10    | <input type="radio"/> |
| <b>D</b> | 2     | 7     | 10    | <input type="radio"/> |

Turn over for the next question

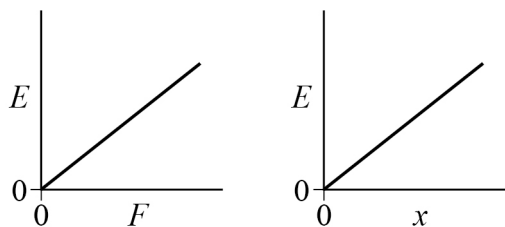


**1 9** A wire obeys Hooke's law. When an applied force  $F$  acts on the wire, the extension is  $x$  and the energy stored in the wire is  $E$ .

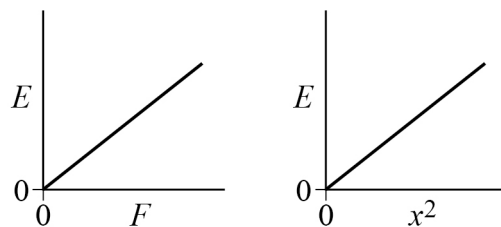
Which pair of graphs is correct?

[1 mark]

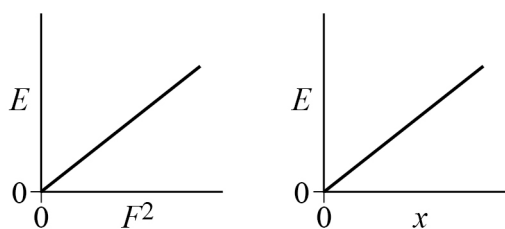
**A**



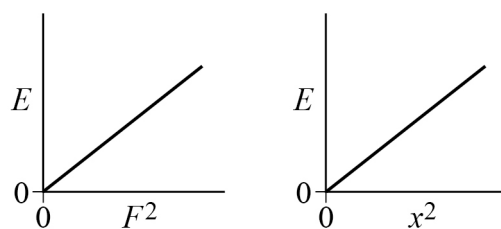
**B**



**C**



**D**



**A** ☐

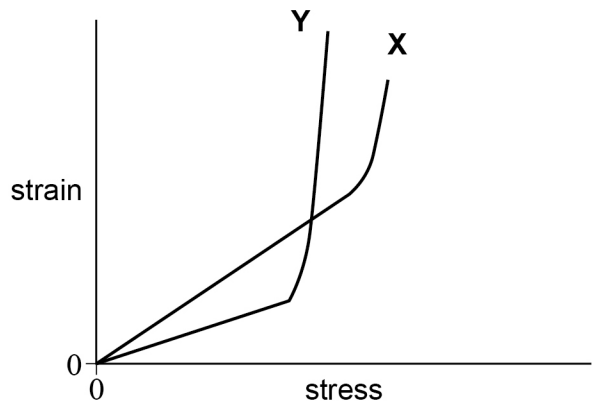
**B** ☐

**C** ☐

**D** ☐



The graph shows the variation of strain with stress for two materials **X** and **Y** up to the points at which they break.



Which row is correct?

[1 mark]

|   | Material with greater<br>Young modulus | Material with greater<br>breaking stress |
|---|----------------------------------------|------------------------------------------|
| A | X                                      | X                                        |
| B | X                                      | Y                                        |
| C | Y                                      | X                                        |
| D | Y                                      | Y                                        |

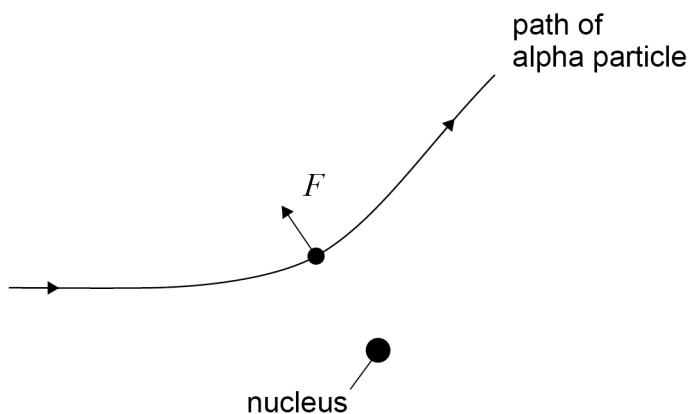
☐☐☐☐

Turn over for the next question



2 1

A nucleus of lead deflects an alpha particle with a force  $F$ .



At the instant shown in the diagram, the force exerted by the alpha particle on the lead nucleus has a magnitude that is

[1 mark]

**A** zero.

☐

**B** less than the magnitude of  $F$ .

☐

**C** equal to the magnitude of  $F$ .

☐

**D** greater than the magnitude of  $F$ .

☐


2 2

When each nucleus in a radioactive source decays,  $8.8 \times 10^{-13}$  J of energy is released.

The source has a half-life of 90 years and an initial activity of 60 MBq.

What power will be available from the source after 270 years?

[1 mark]

A  $6.6 \times 10^{-7}$  W

☐

B  $6.6 \times 10^{-6}$  W

☐

C  $1.3 \times 10^{-5}$  W

☐

D  $1.8 \times 10^{-5}$  W

☐

2 3

A nuclide decays through a radioactive decay chain.

Four  $\beta^-$  particles and two  $\alpha$  particles are emitted.

The final nuclide

[1 mark]

A has a higher proton number than the original nuclide.

☐

B has the same nucleon number as the original nuclide.

☐

C has the same proton number as the original nuclide.

☐

D has a higher nucleon number than the original nuclide.

☐

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

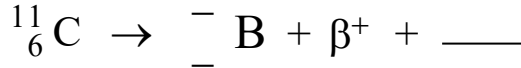
Copyright © 2022 Oxford International AQA Examinations and its licensors. All rights reserved.



**0****2**

Complete the equation for the beta-plus ( $\beta^+$ ) decay of a carbon-11 nucleus.

**[2 marks]**



---

**2**

|   |   |   |   |
|---|---|---|---|
| 0 | 8 | . | 1 |
|---|---|---|---|

Calculate the specific charge of a nucleus of gold-197  $\left({}^{197}_{79}\text{Au}\right)$ .

State an appropriate unit for your answer.

**[3 marks]**

specific charge = \_\_\_\_\_ unit \_\_\_\_\_

**Question 8 continues on the next page**

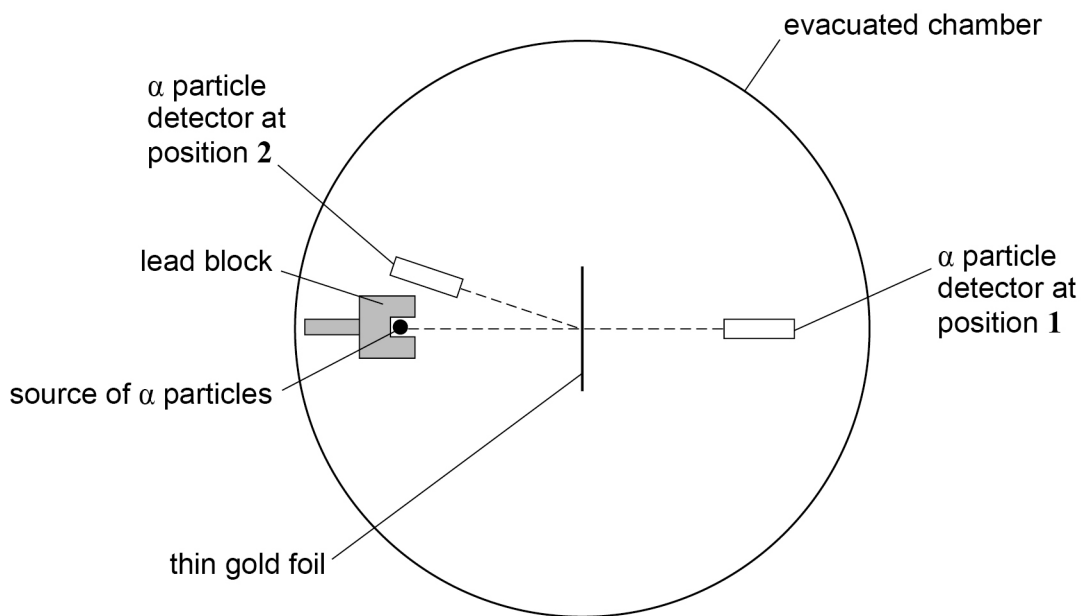
Turn over ►



**Figure 7** shows a Rutherford scattering experiment in which alpha ( $\alpha$ ) particles are directed at a thin gold foil.

A detector counts the number of alpha particles scattered by the thin gold foil per second.

**Figure 7**



0 8 . 2

**Figure 7** shows two positions of the detector in the evacuated chamber.

Compare the numbers of alpha particles per second detected at position **1** and position **2**.

[1 mark]

---



---

0 8 . 3

State **two** conclusions from the Rutherford scattering experiment about the structure of the atom.

[2 marks]

1 

---

---

---

2 

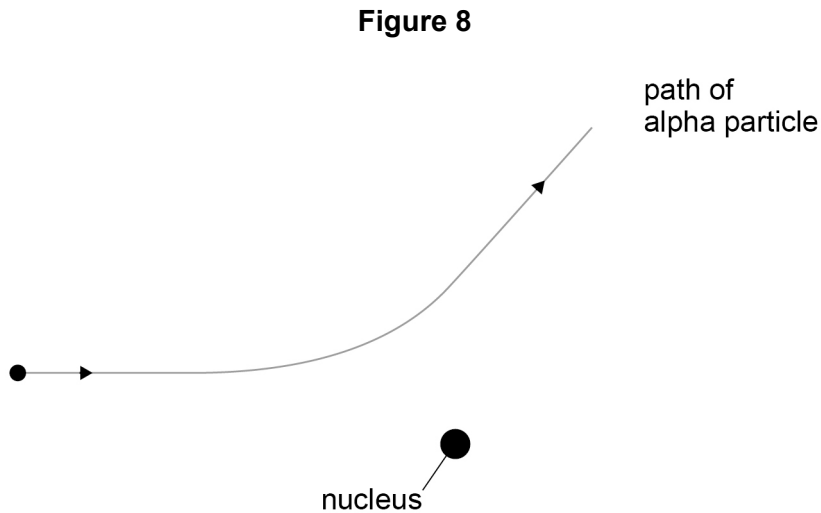
---

---

---



**Figure 8** shows the path of an alpha particle moving close to a gold nucleus.



The experiment is repeated with thin copper-63 ( ${}^{63}_{29}\text{Cu}$ ) foil instead of gold.

**08.4**

Sketch, on **Figure 8**, a line showing the path taken by an alpha particle with the same energy and starting from the same initial position.

[1 mark]

**08.5**

Explain your answer to Question **08.4**.

[2 marks]

---

---

---

---

---

---

---

---

**END OF SECTION A**



## Section B

Answer **all** questions in this section.

*Do not write  
outside the  
box*

09

Technetium-99m is a radioactive isotope used in hospitals.

09

1

Identify the type of radiation that is emitted when a technetium-99m nucleus de-excites.

[1 mark]

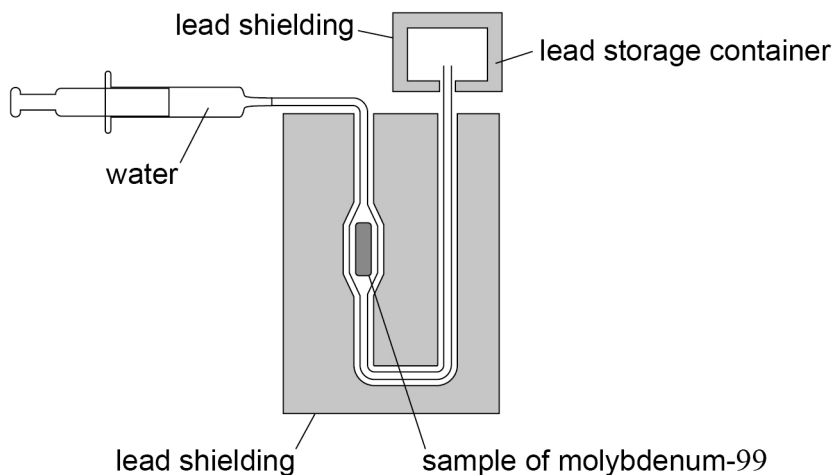
09

2

Samples of technetium-99m are prepared at a hospital from a source of the radioactive isotope molybdenum-99 that is stored at the hospital.

**Figure 9** shows the generator that is used to prepare the sample of technetium-99m.

**Figure 9**



Water from a syringe passes through the molybdenum-99 sample.

The technetium-99m that forms from the decay of the molybdenum-99 dissolves in the water. The water containing technetium-99m is collected in a lead storage container.

Explain how the apparatus in **Figure 9** allows safe handling of the collected sample.

[2 marks]

---

---

---

---

---

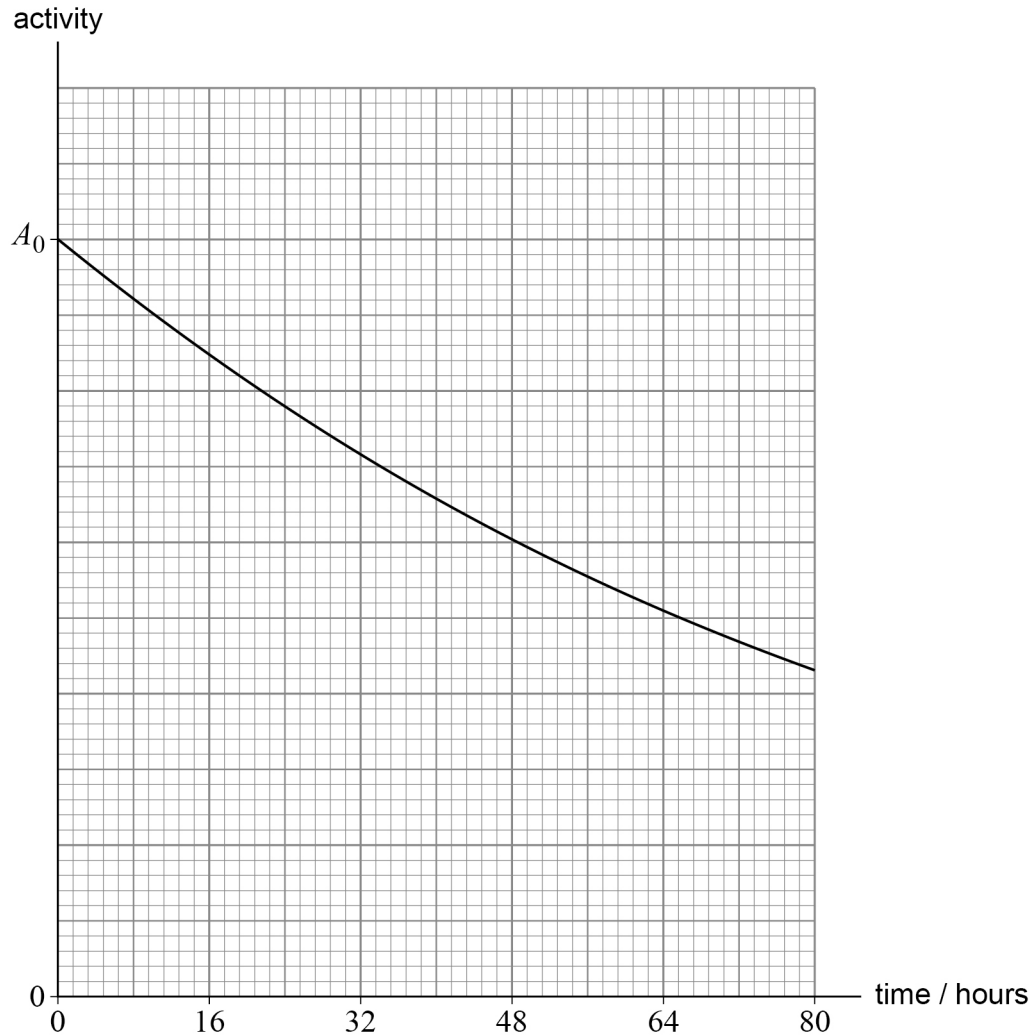


0 9 . 3

**Figure 10** shows the variation of the activity of molybdenum-99 in the generator over an 80-hour period.

$A_0$  is the initial activity of the molybdenum-99 in the generator.

**Figure 10**



Estimate the half-life of molybdenum-99.

**[1 mark]**

half-life = \_\_\_\_\_ hours

**Question 9 continues on the next page**

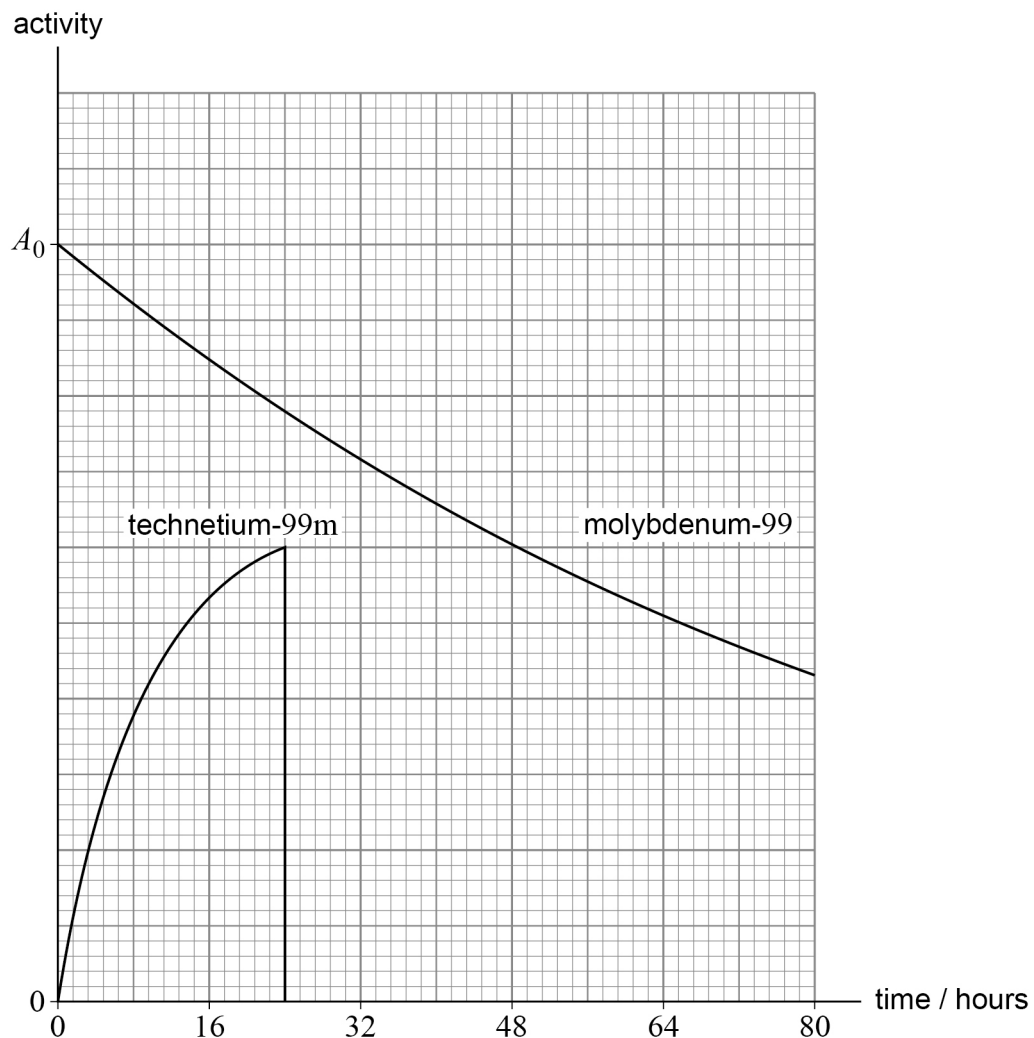
**Turn over ►**



0 9 . 4

**Figure 11** shows the activity of the molybdenum-99 over an 80-hour period and the activity of the technetium-99m that is formed in the generator over the first 24 hours of the same period.

**Figure 11**



A sample of technetium-99m is collected from the generator every 24 hours causing the activity of the technetium-99m in the generator to fall to zero.

Sketch, on **Figure 11**, the activity of the **technetium-99m** between 24 and 72 hours.

**[2 marks]**



0 9 . 5

Explain why a sample of technetium-99m is collected every 24 hours and not more frequently.

[2 marks]

---

---

---

---

---

---

8

Turn over for the next question

Turn over ►



Do not write Do not write

**2****1**

Alpha particles

**[1 mark]****A** can be deflected by a strong magnetic field.☐**B** are less ionising than beta particles.☐**C** have a greater specific charge than beta particles.☐**D** obey the inverse-square law in air.☐**Turn over for the next question****Turn over ►**

Do not write

A radioactive source has a half-life of 3 days.

Initially a detector at a fixed distance from the source records an average count rate of 180 counts per minute.

The average background count rate is 20 counts per minute.

What is the average count rate recorded by the detector 6 days later?

**[1 mark]**

**A** 40 counts per minute

☐

**B** 45 counts per minute

☐

**C** 60 counts per minute

☐

**D** 65 counts per minute

☐

A detector is placed 30 cm from a gamma source and records an average count rate of 290 counts per minute.

A second identical detector is placed a distance of 90 cm from the source.  
The average background count rate is 20 counts per minute.

What is the average count rate measured by the second detector?

[1 mark]

**A** 30 counts per minute

☐

**B** 32 counts per minute

☐

**C** 50 counts per minute

☐

**D** 97 counts per minute

☐

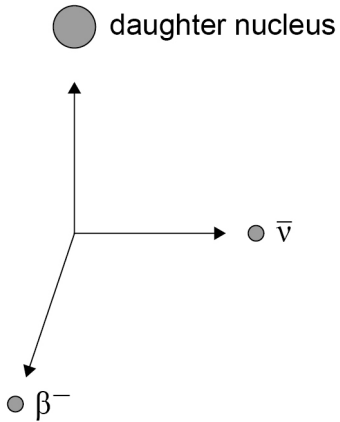
Don't write

Which diagram represents the beta-minus ( $\beta^-$ ) decay of a stationary nucleus with the emission of an antineutrino ( $\bar{\nu}$ )?

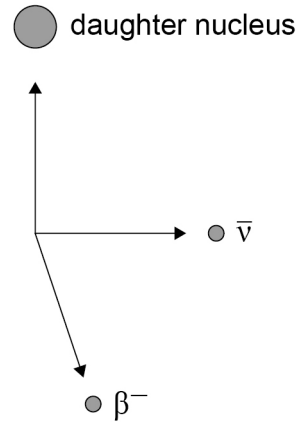
The arrows show the direction of movement of the products of the decay.

[1 mark]

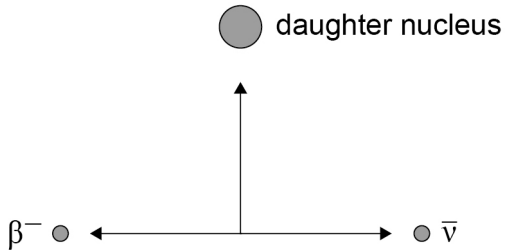
**A**



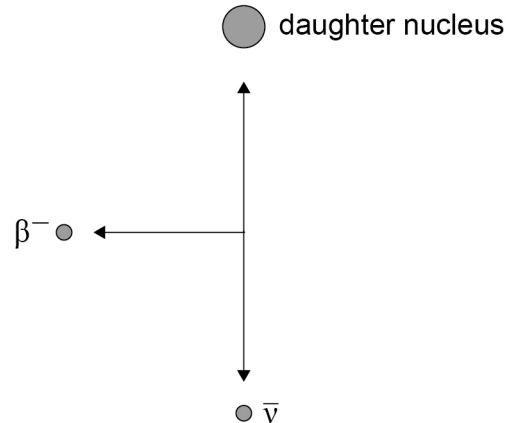
**B**



**C**



**D**



**A**

☐

**B**

☐

**C**

☐

**D**

☐

[illegible]

[illegible]

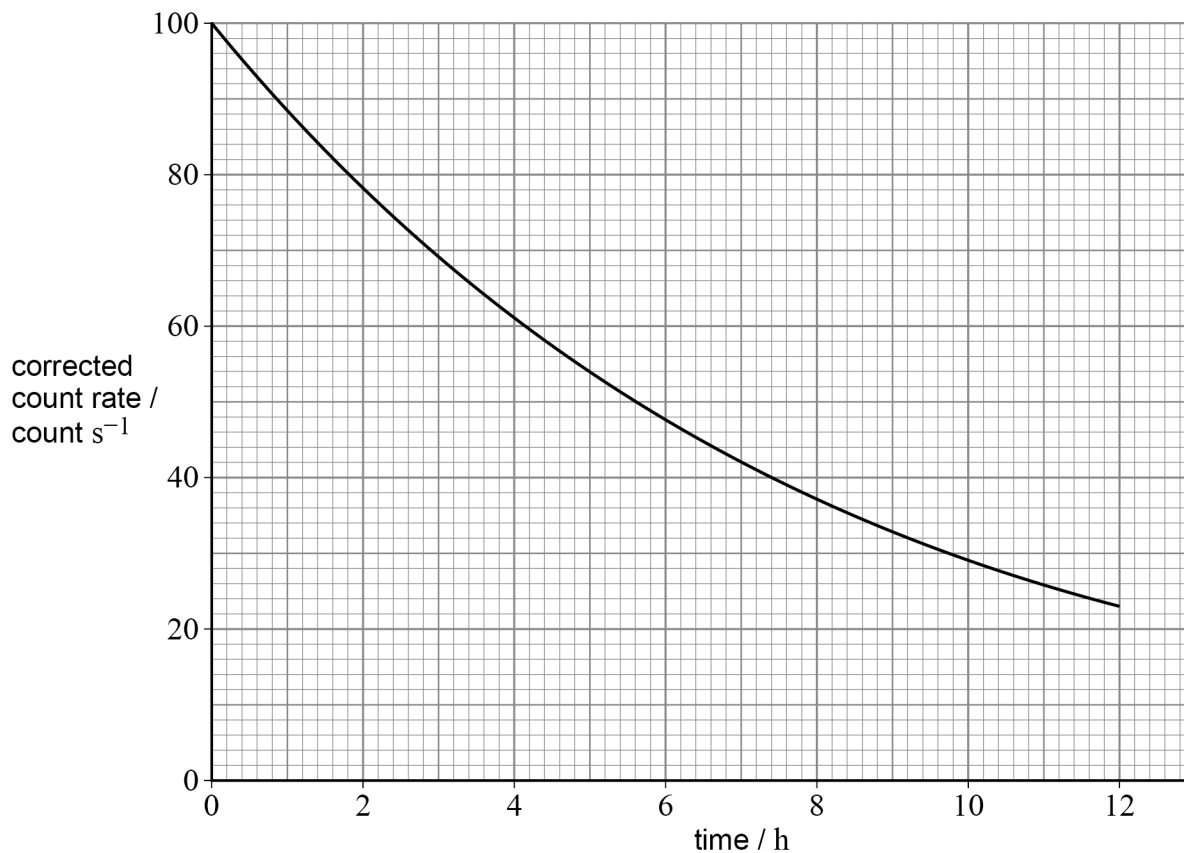
[illegible]

03

A counter is placed near to a radioactive source and the count rate is measured regularly over a 12-hour period.

**Figure 2** shows the variation of corrected count rate with time.

**Figure 2**



03

.1

State what is meant by corrected count rate.

[1 mark]

---

---



0 3 . 2

Determine, using **Figure 2**, an accurate value for the half-life of the decay from the source.

[2 marks]

half-life = \_\_\_\_\_ h

3

Turn over for the next question

Turn over ►



Do not write Do not write

|   |   |
|---|---|
| 0 | 4 |
|---|---|

In the Rutherford scattering experiment, alpha particles were directed at a thin gold foil.

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 1 |
|---|---|---|---|

Explain why the experiment was performed in a vacuum.

[1 mark]

---

---

---

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 2 |
|---|---|---|---|

State how the experimenters detected the alpha particles.

[1 mark]

---

---

---

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 3 |
|---|---|---|---|

Describe the measurements that were made in the Rutherford scattering experiment.

[1 mark]

---

---

---



|                         |                         |
|-------------------------|-------------------------|
| <div>Do not write</div> | <div>Do not write</div> |
|-------------------------|-------------------------|

|   |   |
|---|---|
| 0 | 5 |
|---|---|

A beta-minus ( $\beta^-$ ) particle is emitted when a nucleus of strontium-90 ( ${}^{90}_{38}\text{Sr}$ ) decays into a nucleus of yttrium (Y).

|   |   |   |   |
|---|---|---|---|
| 0 | 5 | . | 1 |
|---|---|---|---|

Explain the origin of the  $\beta^-$  particle.

[2 marks]

---

---

---

---

---

|   |   |   |   |
|---|---|---|---|
| 0 | 5 | . | 2 |
|---|---|---|---|

State the values of  $A$  and  $Z$  for the yttrium nucleus.

[1 mark]

$A =$  \_\_\_\_\_

$Z =$  \_\_\_\_\_

|   |   |   |   |
|---|---|---|---|
| 0 | 5 | . | 3 |
|---|---|---|---|

In early studies of beta decay, the kinetic energies of the  $\beta^-$  particles were measured.

Explain how these measurements improved our understanding of what happens in beta decay.

[2 marks]

---

---

---

---

---

---

---



Do not write Do not write

**2** **2**  $^{208}_{82}\text{Pb}$  can be formed from  $^{232}_{92}\text{U}$  by a series of alpha and beta-minus ( $\beta^-$ ) decays.

How many  $\beta^-$  particles are emitted in the decay series?

**[1 mark]**

**A** 0

**B** 2

**C** 8

**D** 10

Turn over ►





2 3

A radioactive source has an initial activity of  $4.8 \times 10^8$  Bq. The product of the decay is stable.

After exactly 1 day, the activity has fallen to  $1.2 \times 10^7$  Bq.

What is the half-life of the decay?

[1 mark]

**A** 3.2 h

☐

**B** 4.5 h

☐

**C** 5.3 h

☐

**D** 12 h

☐

Technetium-99m is used in medical diagnostics.

Which row shows the emissions from technetium-99m and the reason for its use?

[1 mark]

|   | Emissions      | Reason for use                                                                                            |                       |
|---|----------------|-----------------------------------------------------------------------------------------------------------|-----------------------|
| A | alpha and beta | Alpha and beta emissions have opposite charges that cancel each other out, limiting damage to patients.   | <input type="radio"/> |
| B | beta           | Beta emissions are absorbed by the body so that none can escape from the patient to damage medical staff. | <input type="radio"/> |
| C | gamma          | Gamma emissions are uncharged so will not affect electrical equipment.                                    | <input type="radio"/> |
| D | gamma          | Gamma emissions can escape from the patient and be detected outside.                                      | <input type="radio"/> |

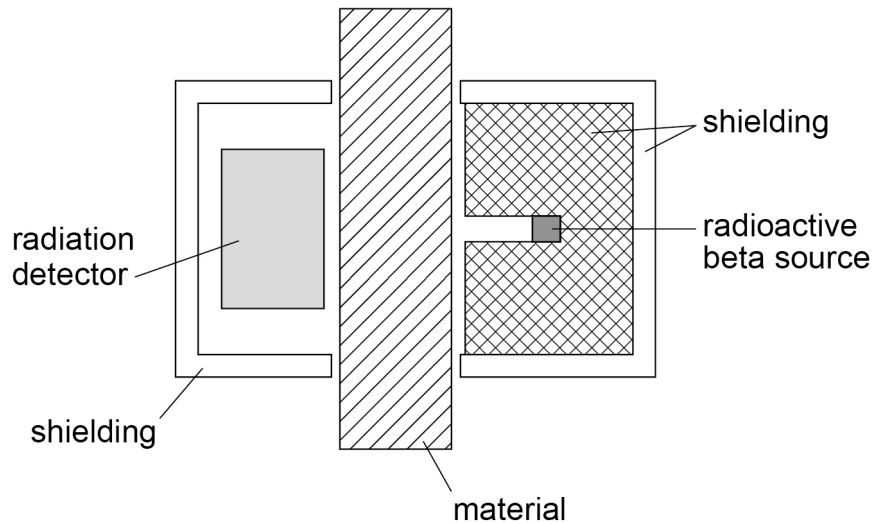
[illegible]

[illegible]



**Figure 3** shows part of a gauge used to measure the thickness of a variety of materials. The gauge uses a radioactive beta source and a radiation detector.

**Figure 3**



05.1

Explain why a gamma source is not suitable for determining the thickness of paper.

**[2 marks]**

---

---

---

---

---

---

**Question 5 continues on the next page**

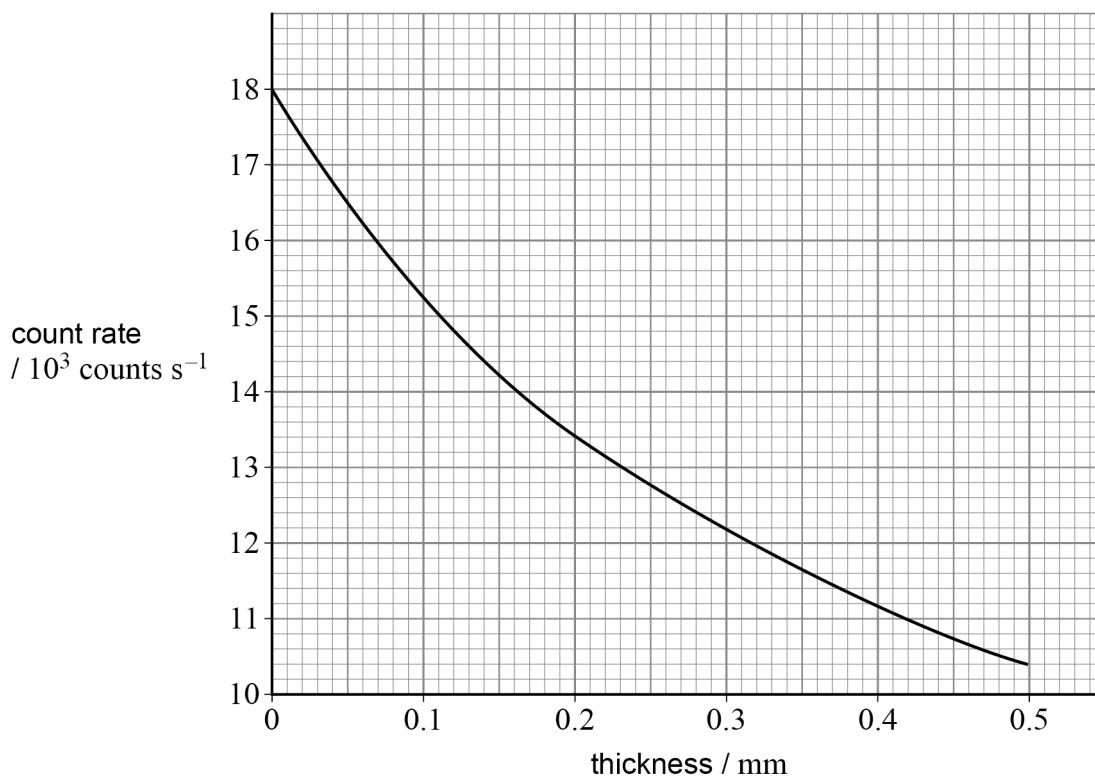


Before the gauge in **Figure 3** can be used, it must be calibrated.  
Count rates are measured for a range of paper thicknesses and a calibration curve is drawn to show the variation of count rate with thickness.

Background radiation is negligible in this situation.

**Figure 4** shows the calibration curve for paper.

**Figure 4**



**0 5 . 2** A paper of thickness 40  $\mu\text{m}$  is tested.

Determine the count rate for this thickness.

**[1 mark]**

count rate = \_\_\_\_\_ counts s<sup>-1</sup>



The beta source in the gauge has a half-life of 3.0 years.

0 5 . 3

Explain, without calculation, whether the same calibration curve can be used after one year.

[2 marks]

---

---

---

---

0 5 . 4

The gauge is used 6.0 years after the calibration shown in **Figure 4**.

Calculate the count rate measured by the detector at this time for paper with a thickness of 0.30 mm.

[2 marks]

count rate = \_\_\_\_\_ counts s<sup>-1</sup>

0 5 . 5

The same gauge is used to determine the thickness of aluminium foil.

The zero-thickness count rate is the same as in **Figure 4**.

Before the gauge is used, a new calibration curve must be produced for aluminium.

Sketch, on **Figure 4**, a curve to show the variation of count rate with thickness for aluminium foil.

[1 mark]

8

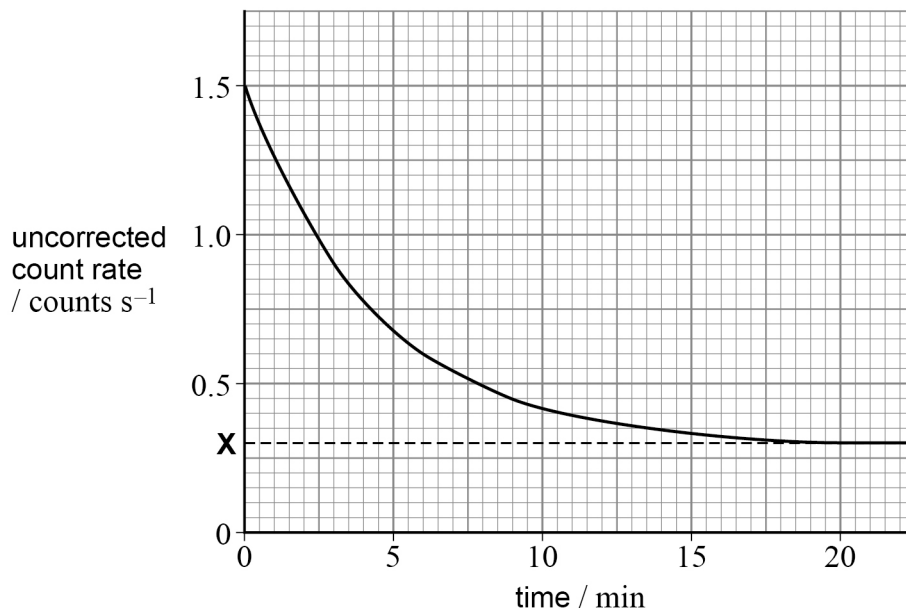


Do not write Do not write

**2 4**

The graph shows the variation of uncorrected count rate with time, recorded by a detector near a radioactive source.

The average background count rate is labelled **X**.



What is the half-life of the radioactive source?

[1 mark]

**A** 3 min

☐

**B** 4 min

☐

**C** 6 min

☐

**D** 19 min

☐

[illegible]

[illegible]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

Copyright © 2023 Oxford International AQA Examinations and its licensors. All rights reserved.



2 3 6 X P H 0

08.1

Describe the simple model of the aluminium-27 ( $^{27}_{13}\text{Al}$ ) atom.

[2 marks]

---

---

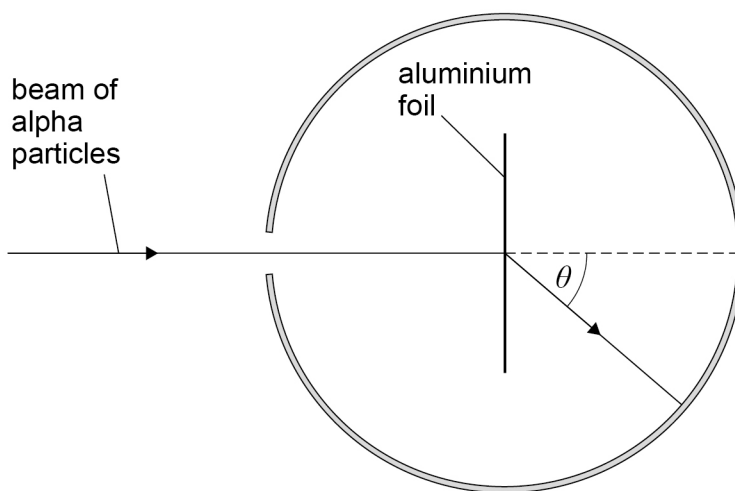
---

---

---

**Figure 5** shows apparatus used in a Rutherford scattering experiment. When a beam of alpha particles was fired at a thin aluminium foil, alpha particles were deflected out of the beam. The experimenters measured the number  $N$  of alpha particles per unit time at various deflection angles  $\theta$ .

**Figure 5**

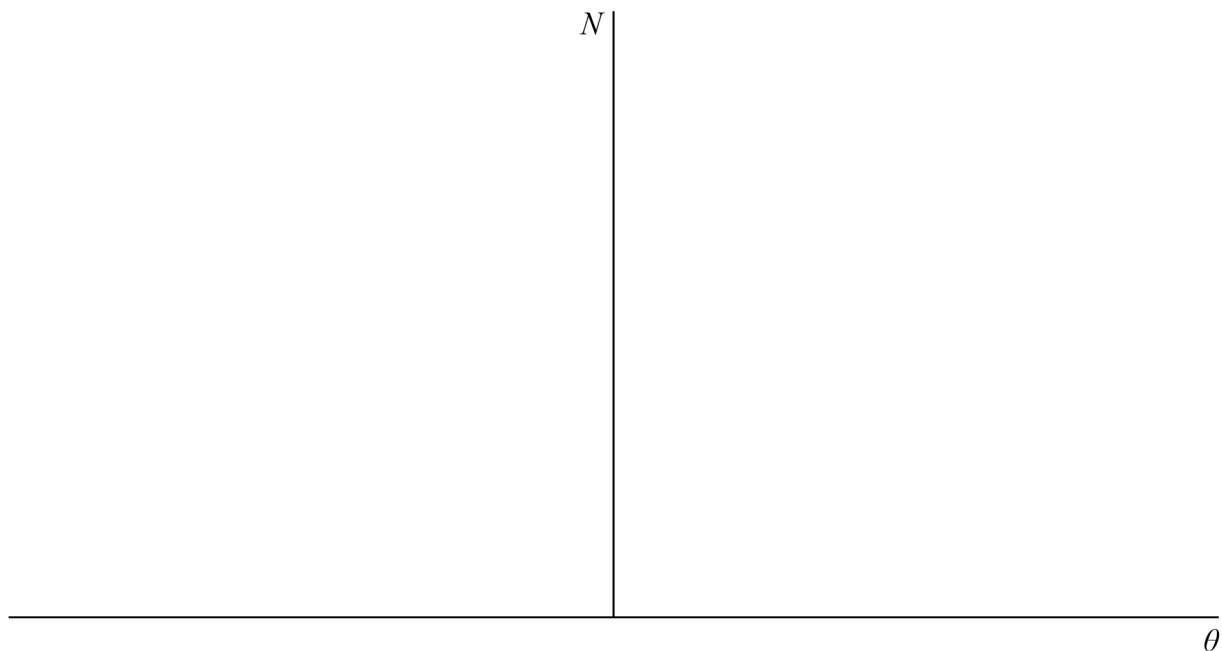


0 8 . 2

Sketch, on **Figure 6**, a graph of the variation of  $N$  with  $\theta$ .  
Label your  $\theta$ -axis with a suitable scale.  
No values are required on the  $N$ -axis.

[2 marks]

**Figure 6**



4

Turn over for the next question

Turn over ►



Do not write

09

A radiation detector is used to measure a background count of 23 counts per minute.

When a gamma source is placed 10.0 cm from the detector, the measured uncorrected count rate is 410 counts per minute.

09.1

Calculate the uncorrected count rate expected when the detector is placed 30.0 cm from the gamma source.

**[3 marks]**

uncorrected count rate = \_\_\_\_\_ counts per minute

A student plans an experiment to verify the inverse-square law for gamma radiation in a school laboratory.

The student writes the following plan. The plan is unsafe and incomplete. The teacher must correct it before the student is allowed to do the experiment.

**Plan**

1. I will hold a gamma source with gloves. My partner will hold a radiation detector.
2. My partner will vary the distance between the detector and the gamma source. The range of distances we will use is 10 cm to 1.0 m, with a measurement taken every 10 cm. He will measure the distances using a metre ruler.
3. We will take readings of the count for one minute at each distance.
4. We will plot a graph of count rate against distance. If this is a straight line, the inverse-square law is verified.





---

---

---

---

---

---

---

---

---



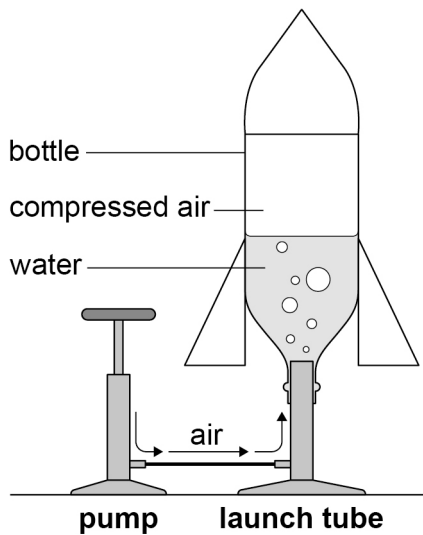
Don't write

A simple rocket is made from a plastic bottle. The rocket is propelled using water and compressed air.

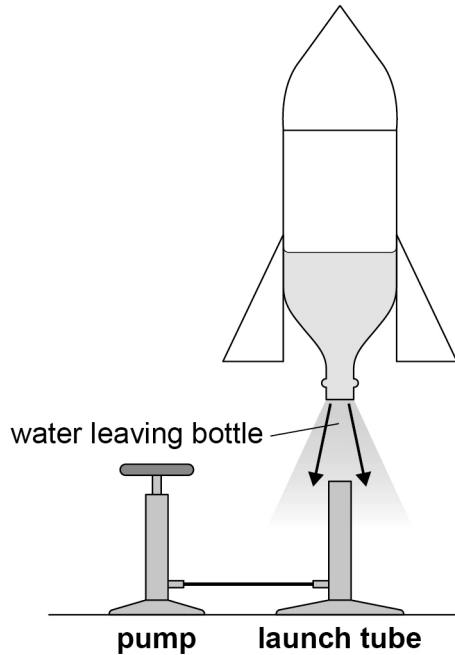
**Figure 7a** shows the rocket being prepared for launch on a launch tube. Additional air is pumped into the bottle through the launch tube.

When the pressure in the bottle is large enough, the bottle is forced off the launch tube. The compressed air pushes the water out of the bottle as shown in **Figure 7b**. The rocket moves vertically upwards.

**Figure 7a**



**Figure 7b**



10.1

Explain why the water leaving the bottom of the bottle causes the rocket to accelerate upwards.

Refer to a relevant physical law in your explanation.

[2 marks]

---

---

---

---

---

---

---

---

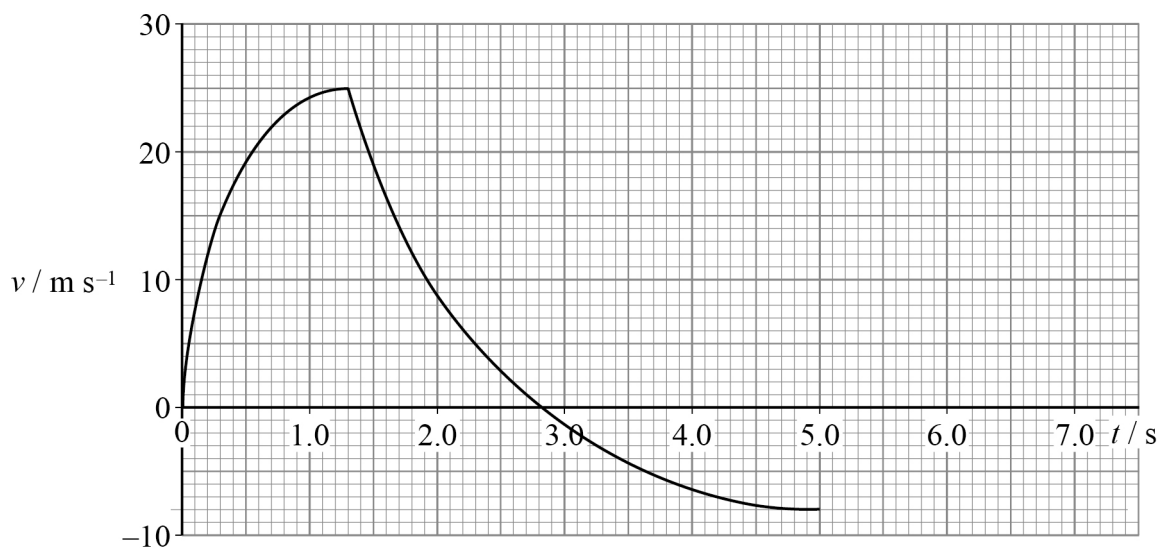
Question 10 continues on the next page

Turn over ►



**Figure 8** shows the variation of vertical velocity  $v$  of the rocket with time  $t$  for the first 5.0 s of its flight.

**Figure 8**



**1 0 . 2** Explain how **Figure 8** shows that there is no water left in the bottle at  $t = 1.3$  s. **[2 marks]**

---



---



---

**1 0 . 3** At  $t = 2.8$  s, the rocket has reached its maximum height.

Determine the maximum height reached by the rocket.

**[3 marks]**

height = \_\_\_\_\_ m



1 0 . 4

Determine the maximum deceleration of the rocket.

[2 marks]

maximum deceleration = \_\_\_\_\_  $\text{m s}^{-2}$

1 0 . 5

Explain why the maximum deceleration of the rocket is greater than  $9.81 \text{ m s}^{-2}$ .

[2 marks]

---

---

---

---

---

---

---

1 0 . 6

The rocket reaches a terminal velocity during its descent, as shown in **Figure 8**.

Explain how you would use **Figure 8** to estimate the total time of flight.

[2 marks]

---

---

---

---

---

---

---

END OF SECTION A



**Section B**

Answer **all** questions in this section.

**1 1 . 1**

State the principle of moments.

**[2 marks]**

---

---

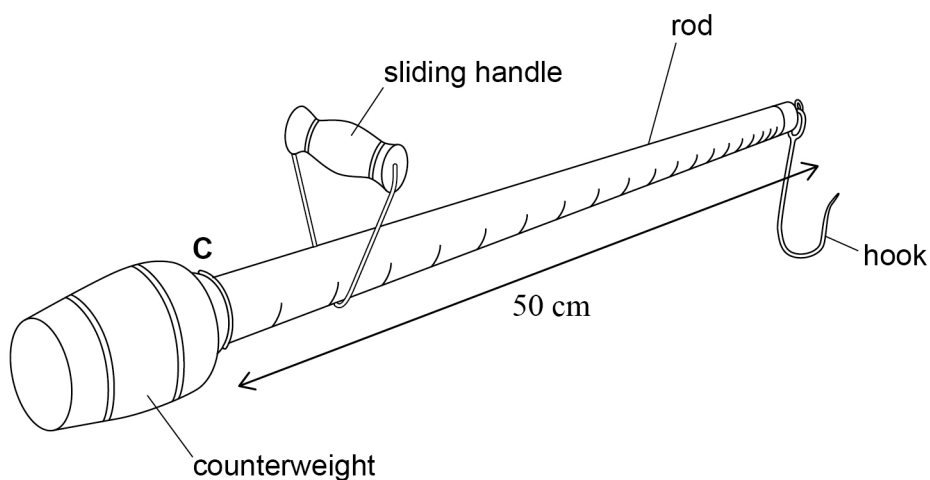
---

---

---

**Figure 9** shows a traditional mass balance called a Bismar scale.

**Figure 9**



A Bismar scale consists of a rod and counterweight of total mass 150 g. The counterweight is fixed to one end of the rod at point **C**. A hook of negligible mass is fixed to the other end of the rod.

When a load is placed on the hook, a sliding handle is moved freely along the rod to balance the Bismar scale in a horizontal position.

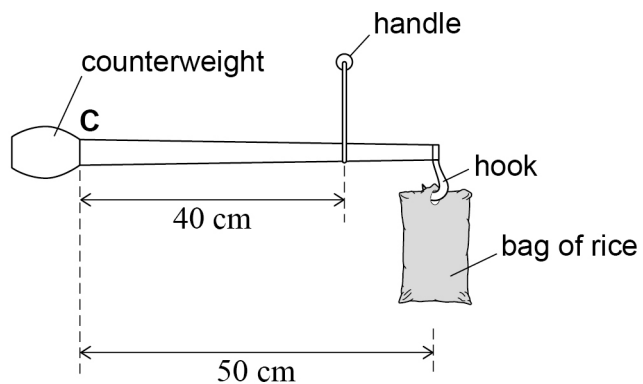
The centre of mass of the Bismar scale is at point **C**. The length of the rod from the hook to point **C** is 50 cm.



1 1 . 2

The Bismar scale is used to determine the mass of a bag of rice.  
The scale is balanced when the sliding handle is 40 cm away from **C**, as shown in **Figure 10**.

**Figure 10**



Calculate the mass of the bag of rice.

**[2 marks]**

mass = \_\_\_\_\_ g

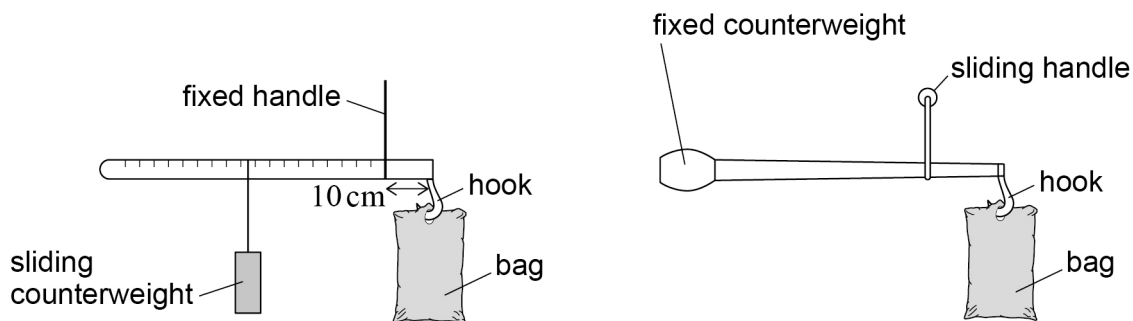
**Question 11 continues on the next page**



1 1 . 3

**Figure 11** shows a simple balance alongside the Bismar scale. The simple balance has a fixed handle and a sliding counterweight. After a bag is placed on the hook, the counterweight is moved to achieve balance.

**Figure 11**



**simple balance**

**Bismar scale**

The simple balance is a uniform rod of length 50 cm and mass 75 g, and a sliding counterweight of mass 75 g.

In the simple balance, the handle is fixed at 10 cm from the hook.

In the Bismar scale, the minimum distance between the sliding handle and the hook is 2 cm.

The simple balance has evenly spaced markings.

The Bismar scale has markings that are closer together as they get closer to the hook.

Compare the advantages and disadvantages of this simple balance and this Bismar scale.

**[3 marks]**



---

---

---

---

---

Turn over for the next question

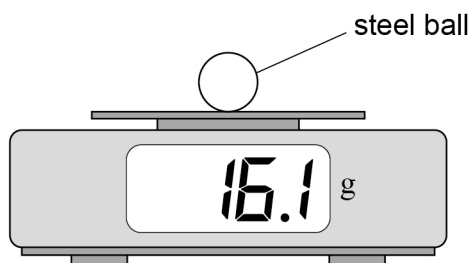


1 2 . 1

A student collects data to determine the density of a steel ball. He finds the mass of the steel ball using a digital balance.

**Figure 12** shows the steel ball on the digital balance.

**Figure 12**



The balance was correctly zeroed before the ball was placed on it.

Calculate the percentage uncertainty in the reading of the mass.

**[1 mark]**

percentage uncertainty = \_\_\_\_\_

1 2 . 2

The student uses a vernier caliper to obtain six measurements for the diameter of the steel ball. The readings are:

15.7 mm      15.5 mm      16.3 mm      15.6 mm      15.7 mm      15.5 mm

Show that the percentage uncertainty in the mean value of the diameter is about 0.6%.

**[3 marks]**



**1 2 . 3** Calculate the density of the steel ball.

**[2 marks]**

density = \_\_\_\_\_  $\text{kg m}^{-3}$

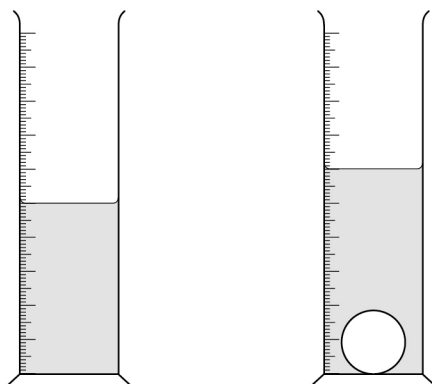
**1 2 . 4** Calculate the percentage uncertainty in the density of the steel ball.

**[1 mark]**

percentage uncertainty = \_\_\_\_\_

**1 2 . 5** When the steel ball is placed into a measuring cylinder containing water, the water level rises as shown in **Figure 13**.

**Figure 13**



Explain **one** reason why the student did not use this method to find the volume of the steel ball.

**[2 marks]**

---



---



---



---

**END OF SECTION B**



## Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS

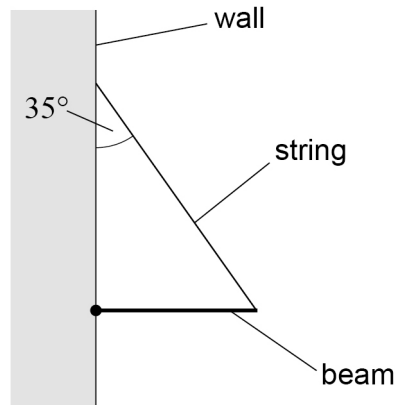


If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional pages for this working.

- 1 3** A uniform beam is attached by a hinge to a vertical wall and supported by a string. The string makes an angle of  $35^\circ$  to the wall.



The tension in the string is 25 N.

What is the weight of the beam?

[1 mark]

**A** 41 N ☐

**B** 29 N ☐

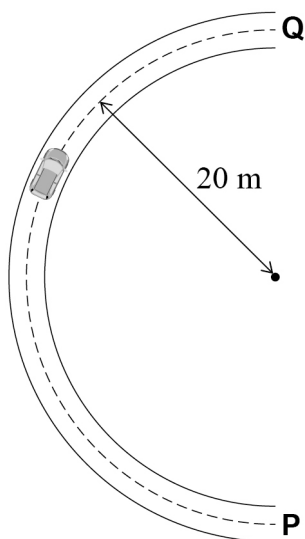
**C** 20 N ☐

**D** 14 N ☐



1 4

A car drives at a constant speed from **P** to **Q** around a semi-circular track of radius 20 m in a time of 4.0 s.



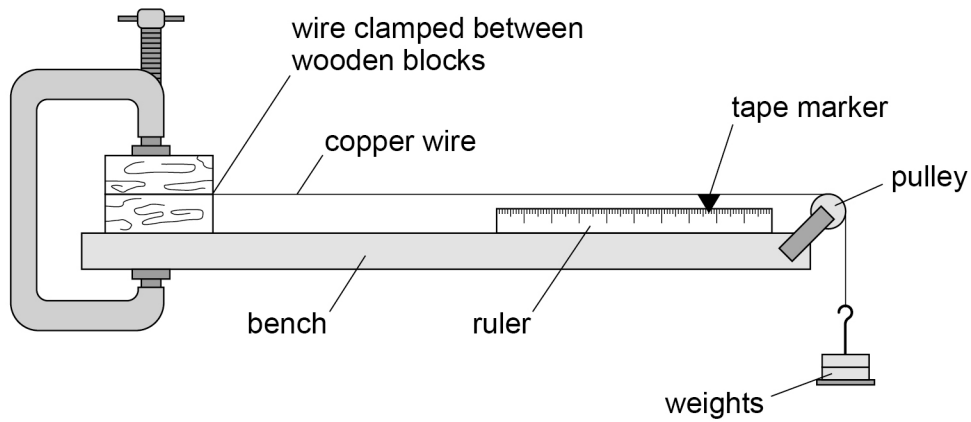
Which row shows the magnitude of the average velocity of the car between **P** and **Q**, and the magnitude of the change in velocity from **P** to **Q**?

[1 mark]

|          | Magnitude of the<br>average velocity / $\text{m s}^{-1}$ | Magnitude of the<br>change in velocity / $\text{m s}^{-1}$ |                       |
|----------|----------------------------------------------------------|------------------------------------------------------------|-----------------------|
| <b>A</b> | 10                                                       | 31                                                         | <input type="radio"/> |
| <b>B</b> | 10                                                       | 0                                                          | <input type="radio"/> |
| <b>C</b> | 16                                                       | 31                                                         | <input type="radio"/> |
| <b>D</b> | 16                                                       | 0                                                          | <input type="radio"/> |



**1 5** The diagram shows equipment that is used in an experiment to determine the Young modulus of copper.



The wire is clamped too loosely.

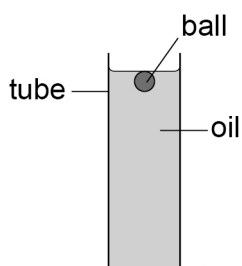
Which row shows the error that occurs due to the loose clamp and the consequence of this error?

[1 mark]

|          | Error                       | Consequence                     |                       |
|----------|-----------------------------|---------------------------------|-----------------------|
| <b>A</b> | extension is overestimated  | Young modulus is underestimated | <input type="radio"/> |
| <b>B</b> | extension is overestimated  | Young modulus is overestimated  | <input type="radio"/> |
| <b>C</b> | extension is underestimated | Young modulus is underestimated | <input type="radio"/> |
| <b>D</b> | extension is underestimated | Young modulus is overestimated  | <input type="radio"/> |



- 1 6** A small steel ball is held below the surface of oil in a tube and released.



Which statement is correct for the ball until it reaches terminal speed?

[1 mark]

- A** The ball's acceleration increases. ☐
- B** The ball's velocity decreases. ☐
- C** The resultant force on the ball increases. ☐
- D** The drag on the ball increases. ☐

- 1 7** A child is ice skating with a skating aid toy.



The child has a mass of 20 kg. The toy has a mass of 15 kg.  
They are initially moving together with a velocity of  $1.5 \text{ m s}^{-1}$ .

The child pushes the toy forwards in the same direction that she is moving.  
The velocity of the toy changes to  $2.0 \text{ m s}^{-1}$ .

What is the speed of the child after she pushes the toy?

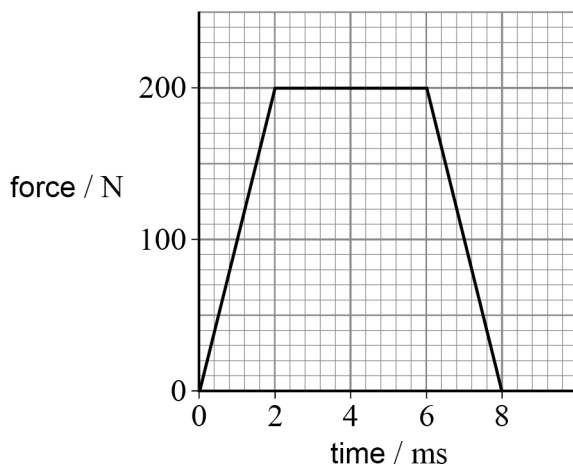
[1 mark]

- A**  $0.83 \text{ m s}^{-1}$  ☐
- B**  $1.0 \text{ m s}^{-1}$  ☐
- C**  $1.1 \text{ m s}^{-1}$  ☐
- D**  $1.3 \text{ m s}^{-1}$  ☐



**1 8**

A ball is hit with a tennis racquet. The graph shows the variation of force on the ball with time.



What is the total impulse exerted on the ball?

**[1 mark]**

**A** 1.6 N s

☐

**B** 1.2 N s

☐

**C** 0.80 N s

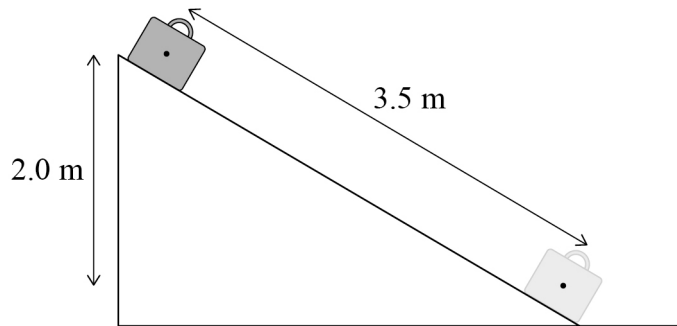
☐

**D** 0.20 N s

☐


**1 9** A suitcase of mass 12 kg slides from rest 3.5 m down a ramp through a vertical height of 2.0 m.

The final speed of the suitcase is  $1.5 \text{ m s}^{-1}$ .



What is the average frictional force between the ramp and the suitcase?

[1 mark]

**A** 220 N ☐

**B** 67 N ☐

**C** 63 N ☐

**D** 14 N ☐

**2 0** A  ${}^4_2\text{He}$  nucleus has a specific charge of  $k$ .

What are the specific charges of a  ${}^{24}_{12}\text{Mg}$  nucleus and a  ${}^{200}_{80}\text{Hg}$  nucleus?

[1 mark]

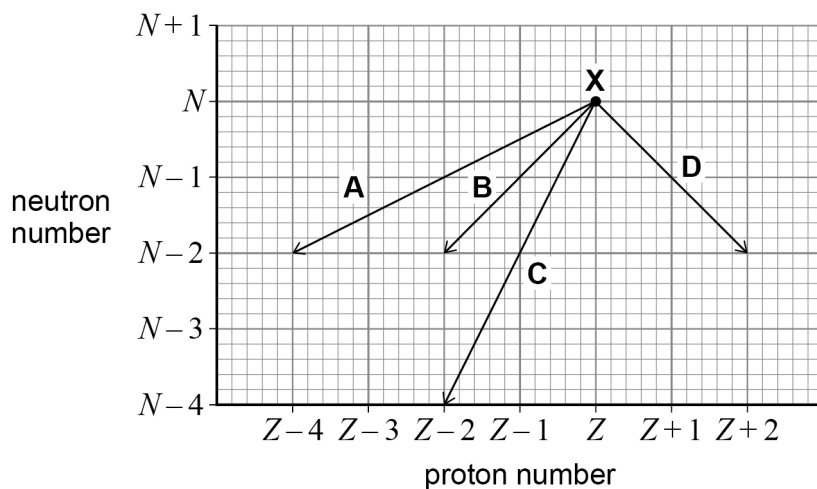
|          | Specific charge of a<br>${}^{24}_{12}\text{Mg}$ nucleus | Specific charge of a<br>${}^{200}_{80}\text{Hg}$ nucleus |                          |
|----------|---------------------------------------------------------|----------------------------------------------------------|--------------------------|
| <b>A</b> | $k$                                                     | $0.4k$                                                   | <input type="checkbox"/> |
| <b>B</b> | $k$                                                     | $0.8k$                                                   | <input type="checkbox"/> |
| <b>C</b> | $6k$                                                    | $0.4k$                                                   | <input type="checkbox"/> |
| <b>D</b> | $6k$                                                    | $0.8k$                                                   | <input type="checkbox"/> |

**2 1** The diagram shows a grid of neutron number against proton number.

Nuclide **X** decays by alpha emission.

Which line represents the change in neutron number and the change in proton number for **X** due to the decay?

**[1 mark]**



**A** ☐

**B** ☐

**C** ☐

**D** ☐



2 2

A photon produces a pair of particles.  
The rest energy of a neutron is approximately 930 MeV.

Which row identifies a pair of particles that could be produced, and the corresponding minimum energy of the photon?

[1 mark]

|          | Pair of particles       | Minimum energy of<br>photon / MeV |                       |
|----------|-------------------------|-----------------------------------|-----------------------|
| <b>A</b> | neutron and antineutron | 930                               | <input type="radio"/> |
| <b>B</b> | neutron and neutron     | 930                               | <input type="radio"/> |
| <b>C</b> | neutron and neutron     | 1860                              | <input type="radio"/> |
| <b>D</b> | neutron and antineutron | 1860                              | <input type="radio"/> |

Turn over for the next question

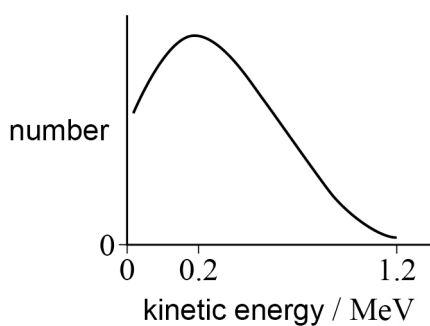
Turn over ►



**2 3**

Bismuth-210 decays to emit a beta particle and an antineutrino.

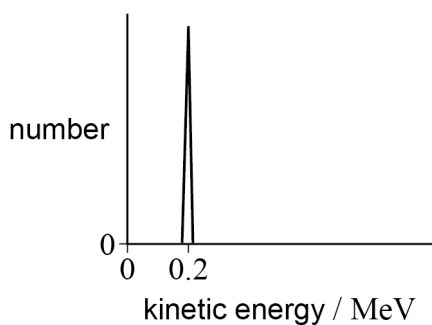
The graph shows the energy distribution of beta particles emitted from the decay of bismuth-210 nuclei.



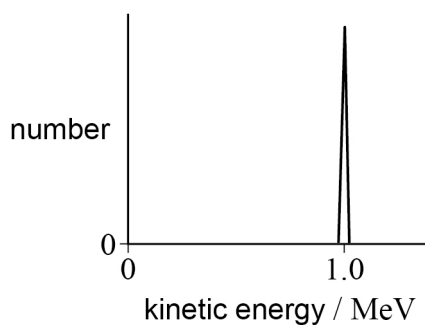
Which graph shows the most likely energy distribution of the antineutrinos emitted?

[1 mark]

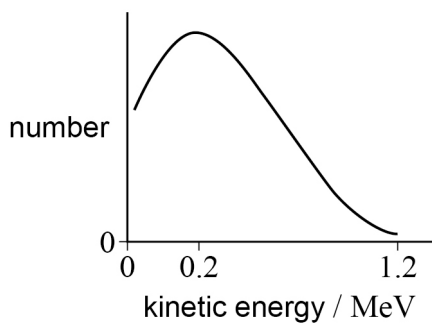
**A**



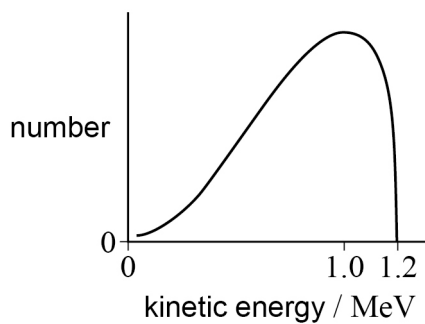
**B**



**C**



**D**



**A**



**B**



**C**



**D**



**2 4** Which beta-decay equation is correct?

[1 mark]

**A**  $^{20}_{8}\text{O} \rightarrow ^{20}_{9}\text{F} + \beta^{-} + \nu_{\text{e}}$  ☐

**B**  $^{20}_{8}\text{O} \rightarrow ^{20}_{9}\text{F} + \beta^{-} + \bar{\nu}_{\text{e}}$  ☐

**C**  $^{20}_{8}\text{O} \rightarrow ^{20}_{9}\text{F} + \beta^{+} + \nu_{\text{e}}$  ☐

**D**  $^{20}_{8}\text{O} \rightarrow ^{20}_{9}\text{F} + \beta^{+} + \bar{\nu}_{\text{e}}$  ☐

**2 5** When time  $t = 6$  days, the activity of a radioactive source is 320 MBq.  
When  $t = 18$  days, the activity is 20 MBq.

What is the activity when  $t = 0$ ?

[1 mark]

**A** 5100 MBq ☐

**B** 1900 MBq ☐

**C** 1300 MBq ☐

**D** 640 MBq ☐

**2 6** A nucleus of barium-137 in an excited state decays to its ground state.

What is the decay process for this nucleus?

[1 mark]

**A** alpha emission ☐

**B** beta emission ☐

**C** gamma emission ☐

**D** neutron emission ☐

[illegible]

[illegible]

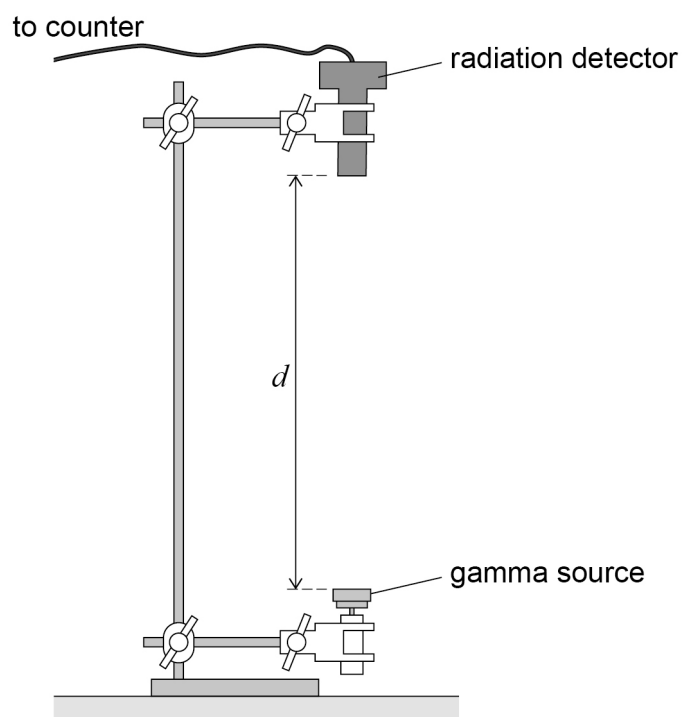
Copyright © 2024 OxfordAQA International Examinations and its licensors. All rights reserved.



03

**Figure 2** shows the apparatus that a student uses to investigate the inverse-square law for gamma radiation. The student uses a source of gamma radiation together with a radiation detector to determine the count rate.

**Figure 2**



The student uses a ruler to measure the vertical distance  $d$  between the **top** of the gamma source and the **bottom** of the radiation detector.

03.1

Explain how measuring  $d$  as shown in **Figure 2** introduces a systematic error.

**[2 marks]**

---



---



---



---



---



0 3 . 2

Describe the steps needed to make the determination of the count rate accurate.

**[2 marks]**

---

---

---

---

4

**Turn over for the next question**

**Turn over ►**



Do not write

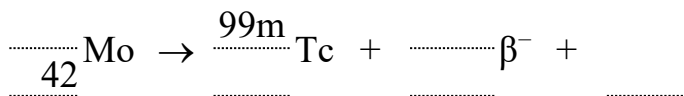
**0 5**

A solution containing a nuclide of technetium-99m is prepared for use in medical diagnosis.

An isotope of molybdenum (Mo) undergoes beta-minus decay to produce a nucleus of technetium-99m (Tc-99m).

**0 5 . 1**

Complete the decay equation.

**[2 marks]****0 5 . 2**

Technetium-99m is metastable.

Explain what is meant by metastable.

**[2 marks]**


---



---



---



---

**0 5 . 3**

The half-life of technetium-99m is 6.0 hours.

A sample of technetium-99m has an initial activity of 384 MBq.

Calculate the time taken for the activity of the sample to decrease to 1.50 MBq.

**[2 marks]**

time taken = \_\_\_\_\_ hours

**Question 5 continues on the next page**

**Turn over ►**



0 5 . 4

A patient is injected with the solution containing technetium-99m and the radiation is detected outside of the body.

Explain why technetium-99m is an appropriate nuclide for use in medical diagnosis.

**[2 marks]**

---

---

---

---

---

---

---

---

8



Don't write

1

2

A nucleus of  ${}^{170}_{74}\text{W}$  decays by alpha emission to form nuclide **X**.

What is the specific charge of the nucleus of **X**?

[1 mark]

**A**  $2.21 \times 10^7 \text{ C kg}^{-1}$

☐

**B**  $3.99 \times 10^7 \text{ C kg}^{-1}$

☐

**C**  $4.18 \times 10^7 \text{ C kg}^{-1}$

☐

**D**  $4.23 \times 10^7 \text{ C kg}^{-1}$

☐

Nuclide **X** is radioactive.

Which decay series produces an isotope of nuclide **X**?

[1 mark]

**A**  $\alpha, \beta^+, \beta^+$

☐

**B**  $\alpha, \beta^-, \beta^-$

☐

**C**  $\alpha, \alpha, \beta^+, \beta^+$

☐

**D**  $\alpha, \alpha, \beta^-, \beta^-$

☐

Turn over for the next question

Turn over ►



Don't write

2

2

What was observed in the Rutherford scattering experiment?

[1 mark]

**A** Few of the alpha particles passed through the gold foil undeflected.

☐

**B** Most alpha particles were deflected through a small angle.

☐

**C** Very few of the alpha particles were deflected through a large angle.

☐

**D** About half of the particles were deflected through angles between  $0^\circ$  and  $90^\circ$ .

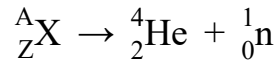
☐

Do not write  
outside the  
box

0 4

Nucleus X decays to form a helium-4 nucleus and a neutron.

The equation for the decay is:



0 4 . 1

Determine the difference between the specific charge of X and the specific charge of the helium-4 nucleus.

[3 marks]

difference in specific charge = \_\_\_\_\_ C kg<sup>-1</sup>

0 4 . 2

X is stationary when the decay occurs.

Compare the motion of the helium-4 nucleus and the motion of the neutron immediately after the decay.

[3 marks]

---

---

---

---

---

---

---

---

---

---



Don't write

|   |   |
|---|---|
| 0 | 6 |
|---|---|

The nuclide americium-241 decays by emitting alpha and gamma radiation.

|   |   |   |   |
|---|---|---|---|
| 0 | 6 | . | 1 |
|---|---|---|---|

Describe an experiment to demonstrate that the intensity of the **gamma** radiation emitted by a point source of americium-241 follows an inverse-square law.

In your answer, describe how to:

- make the necessary measurements
- perform the investigation safely and accurately
- process the measurements to demonstrate the inverse-square law graphically.

You may draw a diagram to help your description.

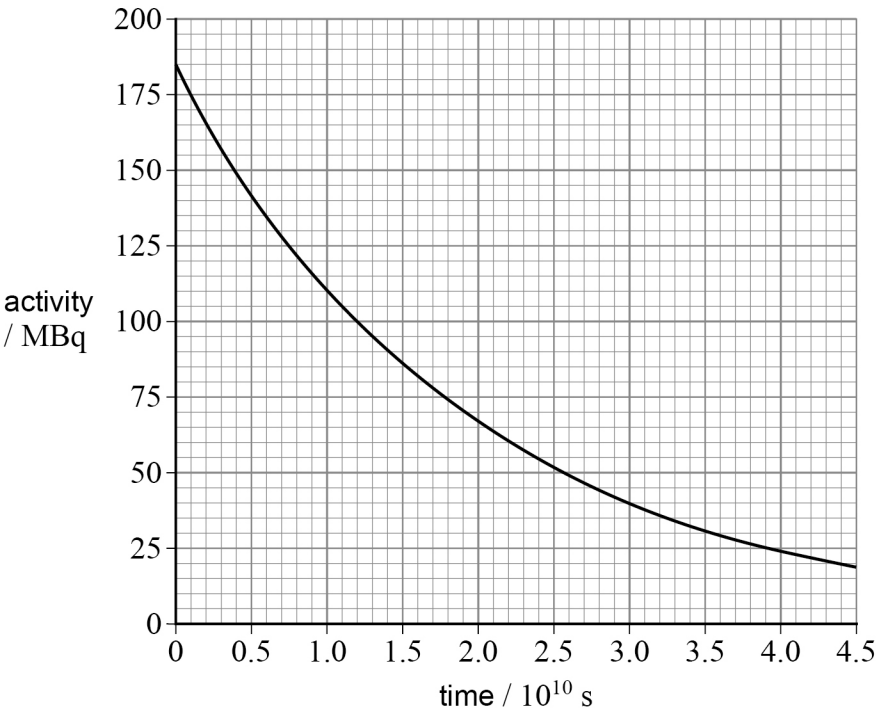
**[5 marks]**

[illegible]

**Figure 5** shows how the activity of an americium-241 source changes with time.

Do not write  
outside the  
box

**Figure 5**



Determine the time taken for the activity to fall to  $\frac{1}{16}$  of the initial value.

**[3 marks]**

time = \_\_\_\_\_ s



Don't write

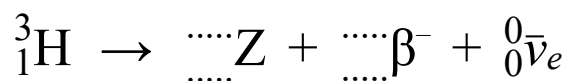
0 9

The nuclide tritium-3 ( ${}^3_1\text{H}$ ) decays to the nuclide Z by  $\beta^-$  emission.

0 9 . 1

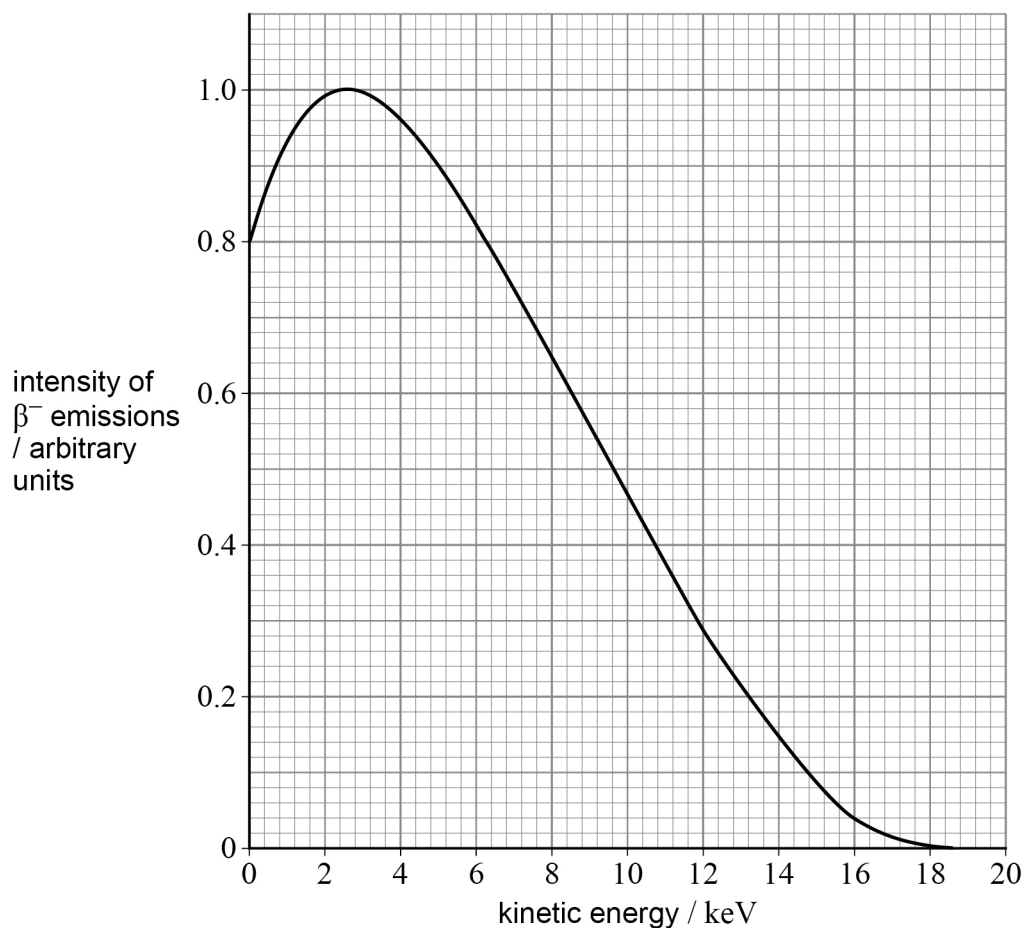
Complete the decay equation for tritium.

[1 mark]



**Figure 9** shows the variation with kinetic energy of the intensity of  $\beta^-$  emissions from tritium-3.

**Figure 9**



0 9 . 2

Determine the speed of a beta-minus particle that is at the peak intensity shown in **Figure 9**.  
Ignore relativistic effects.

[3 marks]

speed = \_\_\_\_\_ m s<sup>-1</sup>

0 9 . 3

Explain how **Figure 9** gives evidence for the existence of the neutrino.

[3 marks]

---

---

---

---

---

---

---

---

0 9 . 4

State the minimum and maximum energies of the neutrinos released in the decay of tritium-3.

[1 mark]

minimum energy = \_\_\_\_\_ keV

maximum energy = \_\_\_\_\_ keV

END OF SECTION A



**There are no questions printed on this page**

*Do not write  
outside the  
box*

**DO NOT WRITE ON THIS PAGE  
ANSWER IN THE SPACES PROVIDED**



## Section B

Answer **all** questions in this section.

*Do not write  
outside the  
box*

A radioactive nuclide  $X$  emits alpha particles.  
The energy from the alpha particles is used to power a device.

Each alpha particle has an initial kinetic energy of  $5.9 \text{ MeV}$ .  
The device requires an input power of  $0.20 \text{ mW}$ .

What is the minimum number of nuclei of  $X$  that must decay every second to power the device?

[1 mark]

**A**  $1.9 \times 10^{-8}$

☐

**B**  $2.1 \times 10^8$

☐

**C**  $3.0 \times 10^{10}$

☐

**D**  $2.1 \times 10^{14}$

☐

Don't write

A radioactive source emits  $\beta^-$  and  $\gamma$  rays only.

The corrected count rate from the source is measured over a long period of time.

Four different sets of absorbers are placed between the source and the detector.

The corrected count rate is measured for each set of absorbers.

The sets of absorbers are:

- a 10 cm air gap only
- a 10 cm air gap and a 4 mm thick aluminium sheet
- a 20 cm air gap only
- a 20 cm air gap and a 4 mm thick aluminium sheet.

Which row shows possible corrected count rates for each set of absorbers?

[1 mark]

|          | Corrected count rates / $\text{s}^{-1}$ |                                   |                    |                                   |                       |
|----------|-----------------------------------------|-----------------------------------|--------------------|-----------------------------------|-----------------------|
|          | 10 cm air gap only                      | 10 cm air gap and aluminium sheet | 20 cm air gap only | 20 cm air gap and aluminium sheet |                       |
| <b>A</b> | 600                                     | 300                               | 150                | 100                               | <input type="radio"/> |
| <b>B</b> | 600                                     | 300                               | 150                | 150                               | <input type="radio"/> |
| <b>C</b> | 600                                     | 300                               | 300                | 150                               | <input type="radio"/> |
| <b>D</b> | 600                                     | 300                               | 225                | 75                                | <input type="radio"/> |

**2****5**

The nuclide  $^{232}_{90}\text{Th}$  decays to the nuclide  $^{208}_{82}\text{Pb}$  by a series of  $\alpha$  decays and  $\beta^-$  decays.

How many  $\beta^-$  decays are there in the decay chain?

**[1 mark]**

**A** 4

☐

**B** 6

☐

**C** 8

☐

**D** 10

☐





[illegible]

Copyright © 2025 OxfordAQA International Examinations and its licensors. All rights reserved.



2 5 1 X P H 0 1

0 4 . 1

A free neutron decays.

State the names of the particles created in the decay.

[1 mark]

---

---

0 4 . 2

Describe what happens when a free neutron collides with an antineutron.

[2 marks]

---

---

---

---

---

---

---



Don't write

07

A teacher uses a radium-226 source to demonstrate the properties of alpha radiation.

07.1

Calculate the specific charge of an alpha particle.

[3 marks]

specific charge = \_\_\_\_\_ C kg<sup>-1</sup>

07.2

Radium-226 has a half-life of 1600 years.

Suggest **one** reason why this half-life means that radium-226 is suitable for use in school laboratories.

[1 mark]

---



---



---

The maximum range of alpha particles in air is about 50 mm.

07.3

Explain why the momentum of an alpha particle decreases as it travels through air.

[1 mark]

---



---



---



The nuclide radium-226:

- emits only alpha radiation
- undergoes a series of decays to become the stable nuclide lead-206.

The teacher places the source at different distances from a detector and records the corrected count rate as shown in **Table 1**.

**Table 1**

| Distance / mm | Corrected count rate / count s <sup>-1</sup> |
|---------------|----------------------------------------------|
| 25            | 4502                                         |
| 75            | 151                                          |

0 7 . 4

Describe a procedure the teacher can use to determine the corrected count rate.

**[2 marks]**

---

---

---

---

---

---

0 7 . 5

Explain why the corrected count rate shown for a distance of 75 mm in **Table 1** is not zero.

**[2 marks]**

---

---

---

---

---

---



Don't write

Carbon-14 has a half-life of approximately 5700 years and can be used to date an old piece of wood.

1.0 g of the wood originally contained  $6.5 \times 10^{10}$  atoms of carbon-14.

A 0.30 g sample of the wood now contains  $2.4 \times 10^9$  atoms of carbon-14.

What is the best estimate for the age of the 0.30 g sample?

[1 mark]

**A** 5700 years

☐

**B** 12 000 years

☐

**C** 17 000 years

☐

**D** 34 000 years

☐

Don't write

1

2

When a technetium-99m nucleus decays, it emits a gamma ray with an energy of 140 keV.

How many nuclei in a sample of technetium-99m must decay in one second to produce a power output of 2.2 nW?

[1 mark]

**A**  $9.8 \times 10^4$

☐

**B**  $9.8 \times 10^7$

☐

**C**  $9.8 \times 10^{10}$

☐

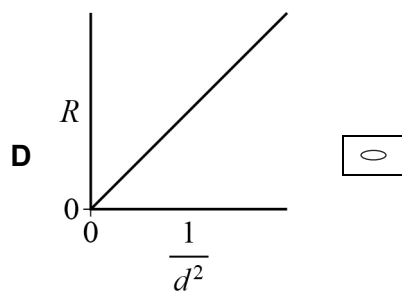
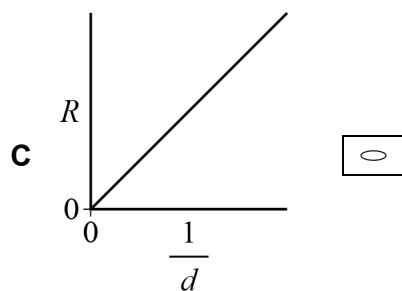
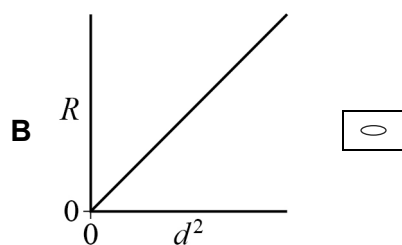
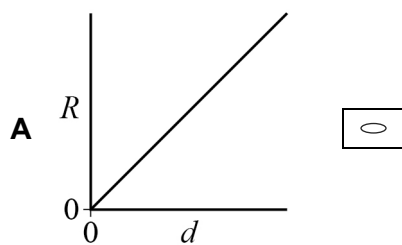
**D**  $9.8 \times 10^{13}$

☐

The corrected count rate  $R$  from a gamma source is measured at different distances  $d$  between the source and a detector.

Which graph is correct?

[1 mark]



Don't write

0 8

A teacher provides a student with a radioactive source and asks the student to identify the type of radiation that it emits.

0 8 . 1

Describe an experiment that the student could do to identify which type of radiation is emitted.

You may wish to include a diagram in your answer.

**[3 marks]**

---

---

---

---

---

---

---

---

**Question 8 continues on the next page**



0 8 . 2

State **two** ways to reduce the hazards of exposure to ionising radiation when using the radioactive source.

[1 mark]

1 \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

2 \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

A radioactive source used in a school contains caesium-137.  
Caesium-137 decays by  $\beta$  emission to form barium-137m.

0 8 . 3

Barium-137m is a metastable isotope with a half-life of 153 s.

Explain what is meant by this statement.

[3 marks]

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

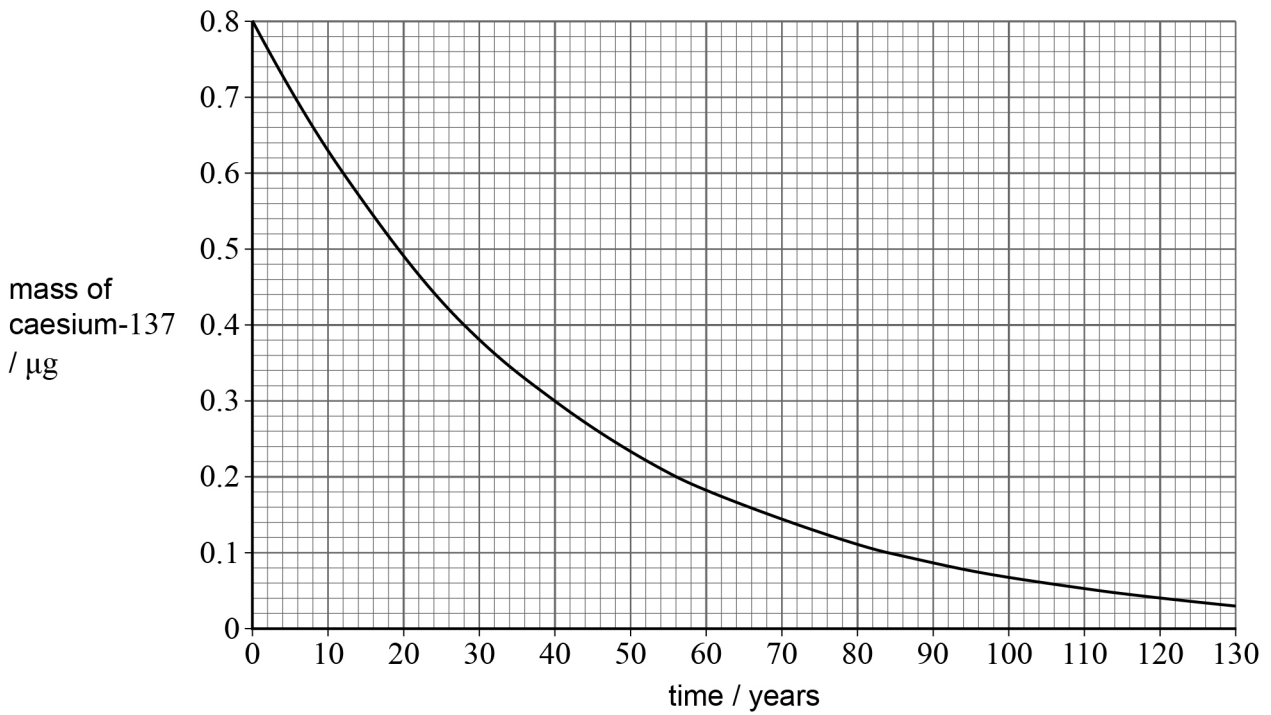
\_\_\_\_\_



0 8 . 4

**Figure 6** shows the variation of the mass of caesium-137 in the source with time.

**Figure 6**



Determine, using **Figure 6**, a value for the half-life of caesium-137 in years.

[1 mark]

half-life = \_\_\_\_\_ years

8

**Turn over for the next question**

**Turn over ►**



Don't write

**1****2**

Which particle has the greatest magnitude of  $\frac{\text{charge}}{\text{mass}}$ ?

**[1 mark]**

**A** alpha particle



**B** electron



**C** neutron



**D** proton



**1****3**

A radioactive nucleus **X** decays into nucleus **Y** by emitting two  $\beta^-$  particles and an  $\alpha$  particle.

Which statement is correct?

**[1 mark]**

**A** **X** and **Y** are isotopes.

☐

**B** **X** and **Y** have the same nucleon number.

☐

**C** **Y** has a larger proton number than **X**.

☐

**D** **Y** has a larger nucleon number than **X**.

☐

Don't write

An average force of 42 kN acts on the air that is passing through a jet engine. This force causes the speed of the air to increase by  $540 \text{ m s}^{-1}$ .

What mass of air passes through the engine in one minute?

[1 mark]

**A**  $7.7 \times 10^{-2} \text{ kg}$

☐

**B**  $4.7 \text{ kg}$

☐

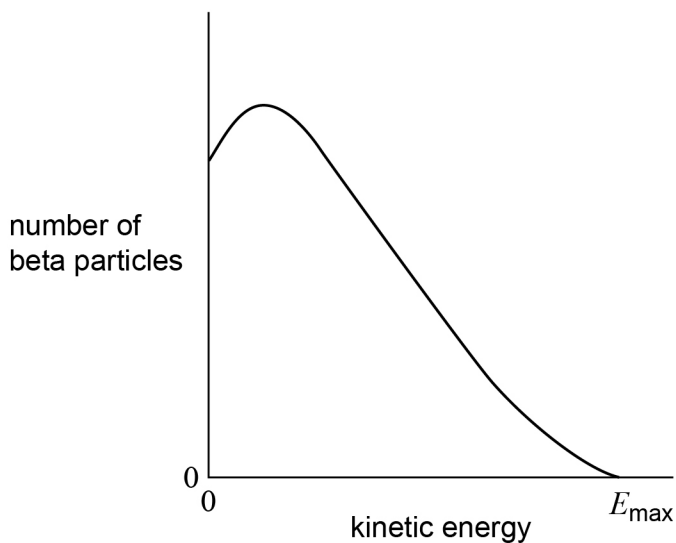
**C**  $78 \text{ kg}$

☐

**D**  $4.7 \times 10^3 \text{ kg}$

☐

The graph shows the distribution of kinetic energies of beta minus particles from the decay of carbon-14.



What can be deduced from this graph?

[1 mark]

- A** A gamma photon is also released to conserve energy.
- B** A neutron turns into a proton.
- C** Each emitted beta particle shares the released energy with another particle.
- D** The emitted beta particles all have the same speed.

☐☐☐☐

**2****4**

The intensity of gamma radiation at a point 45 cm away from a source is  $62 \text{ W m}^{-2}$ .

What is the intensity at a point an additional 75 cm further away from the source?

**[1 mark]**

**A**  $8.7 \text{ W m}^{-2}$

☐

**B**  $22 \text{ W m}^{-2}$

☐

**C**  $23 \text{ W m}^{-2}$

☐

**D**  $37 \text{ W m}^{-2}$

☐

Do not write  
outside the  
box

0 3

A neutron decays by emitting a  $\beta^-$  particle.

0 3 . 1

Identify the other particle that is emitted in the decay.

[1 mark]

---

0 3 . 2

Describe what happens the electron and a positron collide.

[2 marks]

---

---

---

---